Principles of Economics

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- Outline:

1. Introduction
2. Demand and Supply
3. Elasticity
4. Surplus and deadweight loss
5. The obstacles to efficiency

1.Introduction

1.Introduction

- Economics: The study of how society manages its scarce resources.
  - Scarcity
  - Optimal allocation/management of resources
- Economics:
  - Microeconomics
  - Macroeconomics

1.Introduction

- Ten principles of economics:
  - People face trade-offs
  - The cost of something is what you give up to get it
  - Rational people think at the margin
  - People respond to incentives
  - Voluntary trade can make everyone better off
  - Markets are generally a good way to organize economic activity

1.Introduction

- Ten principles of economics (Cont'):
  - Governments can sometimes improve market outcomes
  - The society's standard of living depends on its ability to produce goods and services
  - Prices rise when the government prints too much money
  - Society faces a short-run trade-off between inflation and unemployment.

1.Introduction

- Three key questions in microeconomics:
  - What goods and services should be produced?
  - How should they be produced?
  - Who gets to consume them?

1.Introduction

- Different mechanisms to answer these questions:
  - Government
  - Queue
  - Price (market)
  - Lottery
  - ….

1. Introduction

- The role of economics in identifying the relation between different phenomena:
  - Cause and effect vs. correlation
  - Example:
    - Ancient Russia
    - Correlation between breastfeeding and infant's health
    - Health economics and insurance
    - Free trade and child labor
- Theoretical vs. empirical economics
- Positive vs. normative economics

2.Demand and Supply

2.Demand and Supply

- Key concepts:
  - Market: A mechanism whereby buyers and sellers find each other and buy and sell services
  - Two types of Markets:
    - Factor markets: markets for factors of production
    - Product markets: markets for services and finished goods
  - Capital Markets:
    - Markets where firms raise money for investment by selling debt or equity, as well as the markets where these debt and equity claims are subsequently traded.

- Factors affecting demand for good X:
  - Price of X
    - The higher/lower the price of X, the lower/higher the demand for X (inverse relationship)
  - Income
    - **Normal goods** : direct relationship between income and demand.
      - The higher/lower the income, the higher/lower the demand for X
    - **Inferior goods** : Inverse relationship between income and demand.
      - The higher/lower the income, the lower/higher the demand for X

- Factors affecting demand for good X (Cont'):
  - Price of related goods
    - **Substitute good**: a good that can be used in place of another.
      - The higher/lower the price for substitute good Y, the higher/lower the demand for good X (direct relationship)
      - *Example*: Tea and coffee , hotdog and hamburger
    - **Complementary good**: a good that should be used in conjunction with another good.
      - The higher/lower the price for complementary good Y, the lower/higher the demand for good X (inverse relationship)
      - *Example*: gasoline and automobile , computer and software

2.Demand and Supply

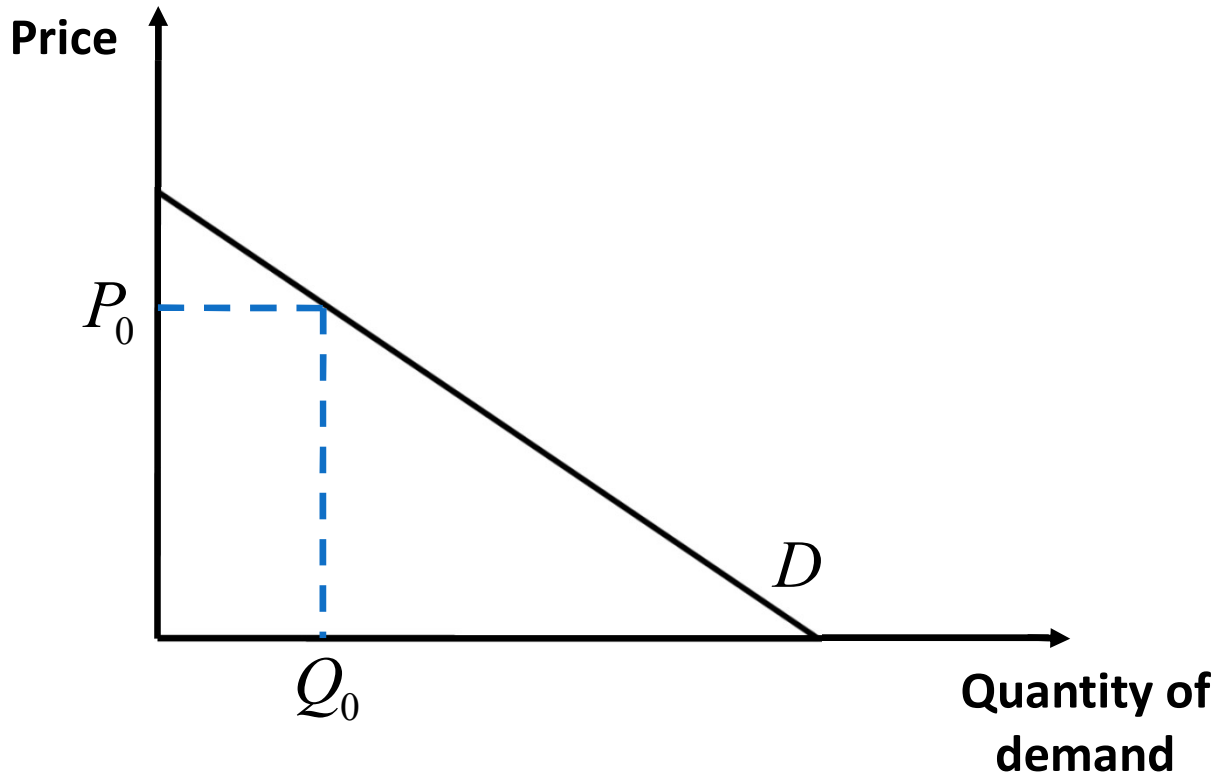
- Demand function:

$$Q_x^D = f(P_x, P_y, I, ...)$$

- Q_x^D : Quantity of demand for x
- P_x : Price for X
- P_y : Price for Y (other goods)
- I : Income

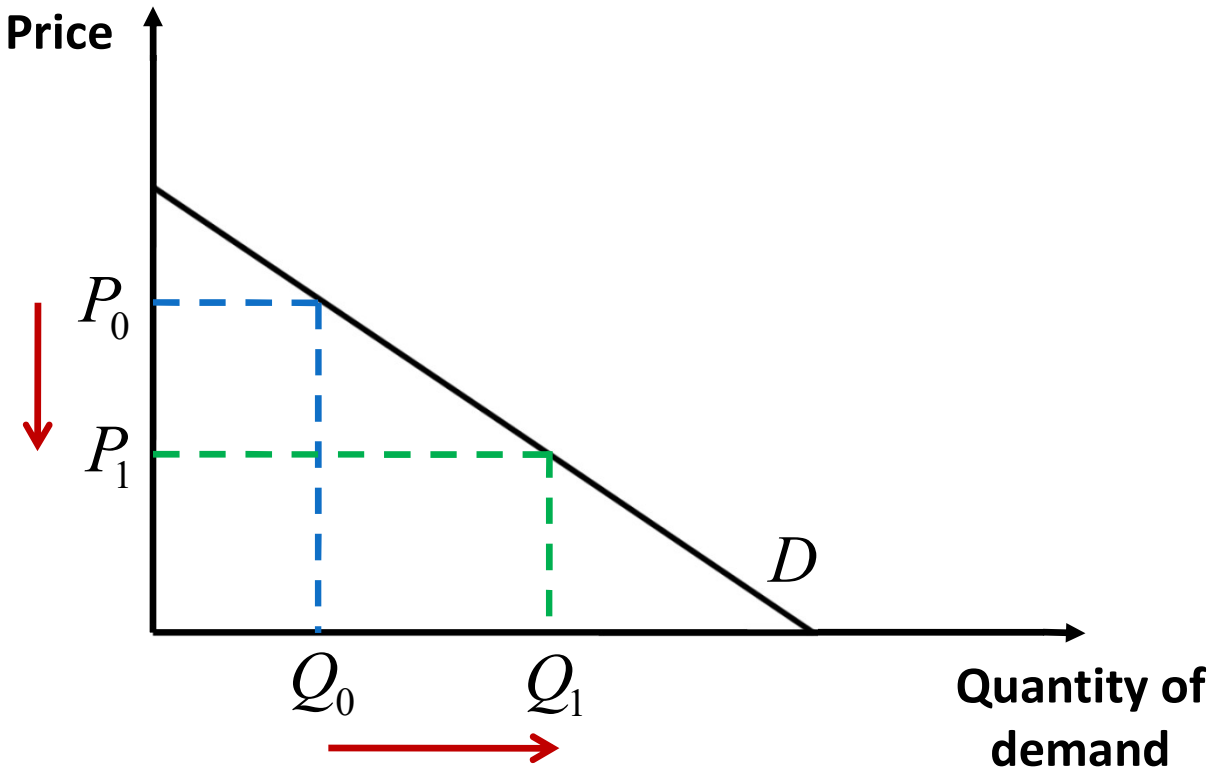
2.Demand and Supply

- Demand Curve: The relationship between price and quantity of demand



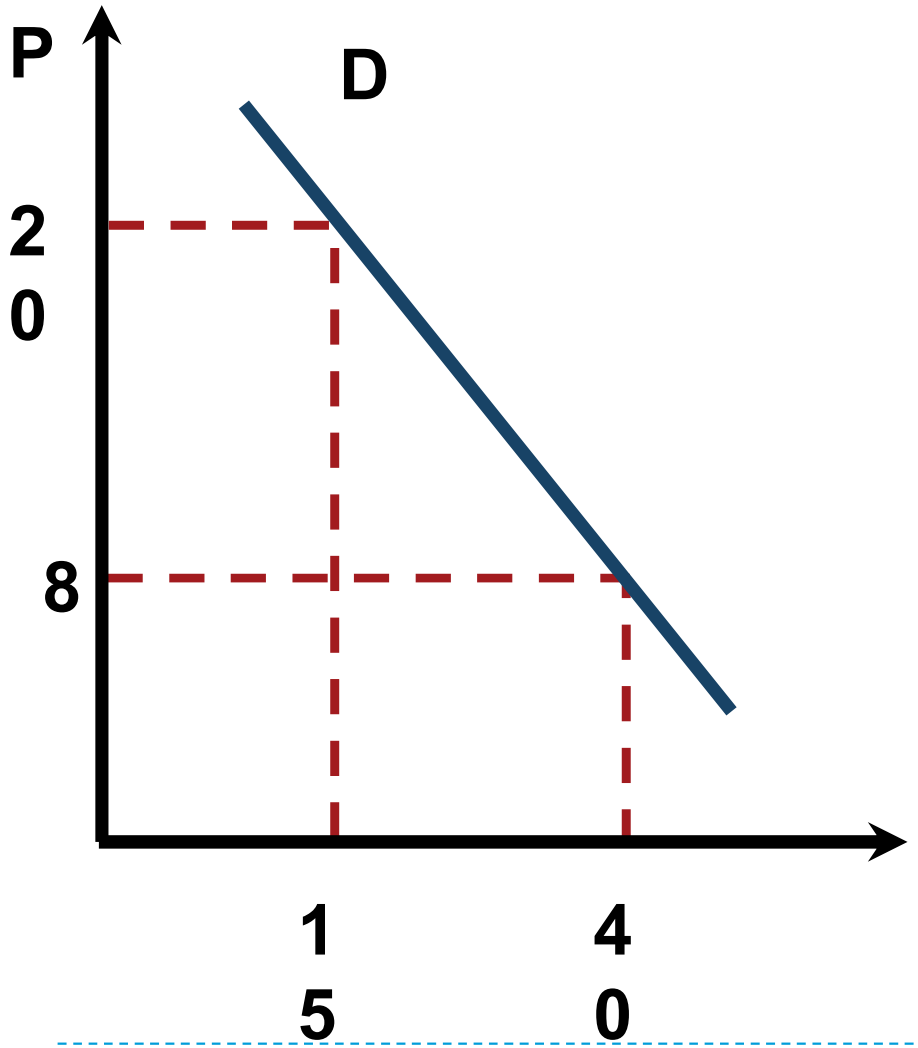
- Two types of changes in demand curve:

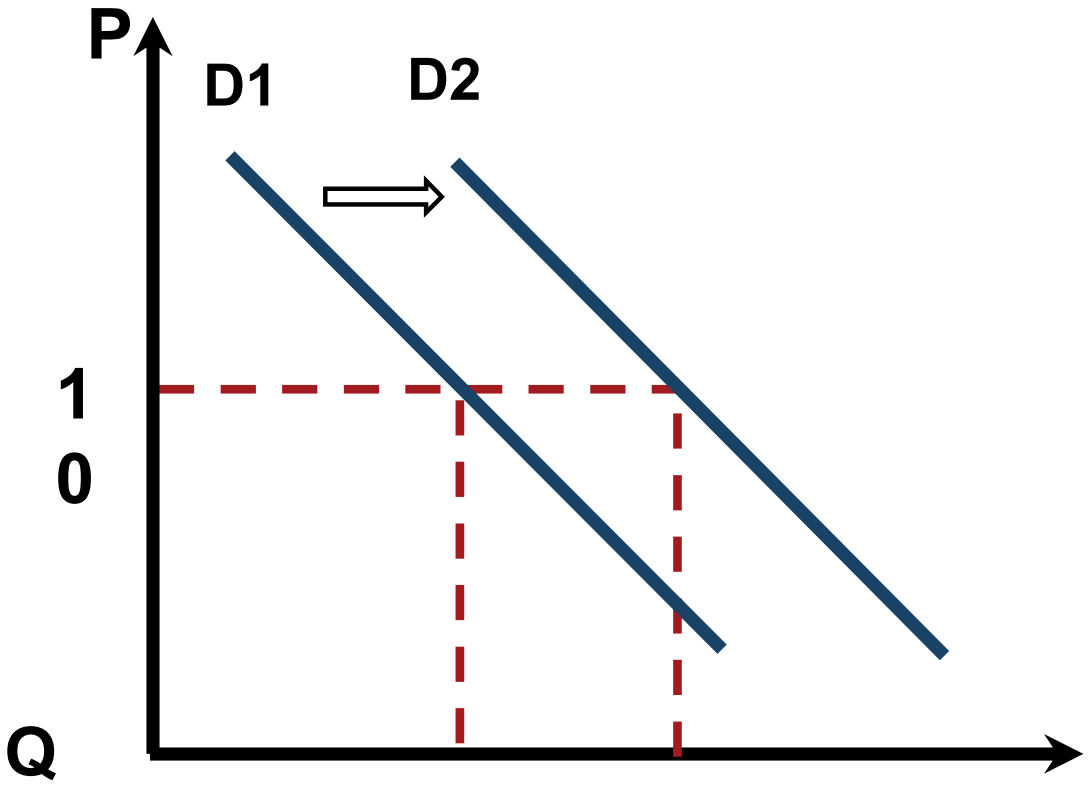
  1.Change along demand curve: when price changes



2.Demand and Supply

2.Shift in demand curve: when other factors change

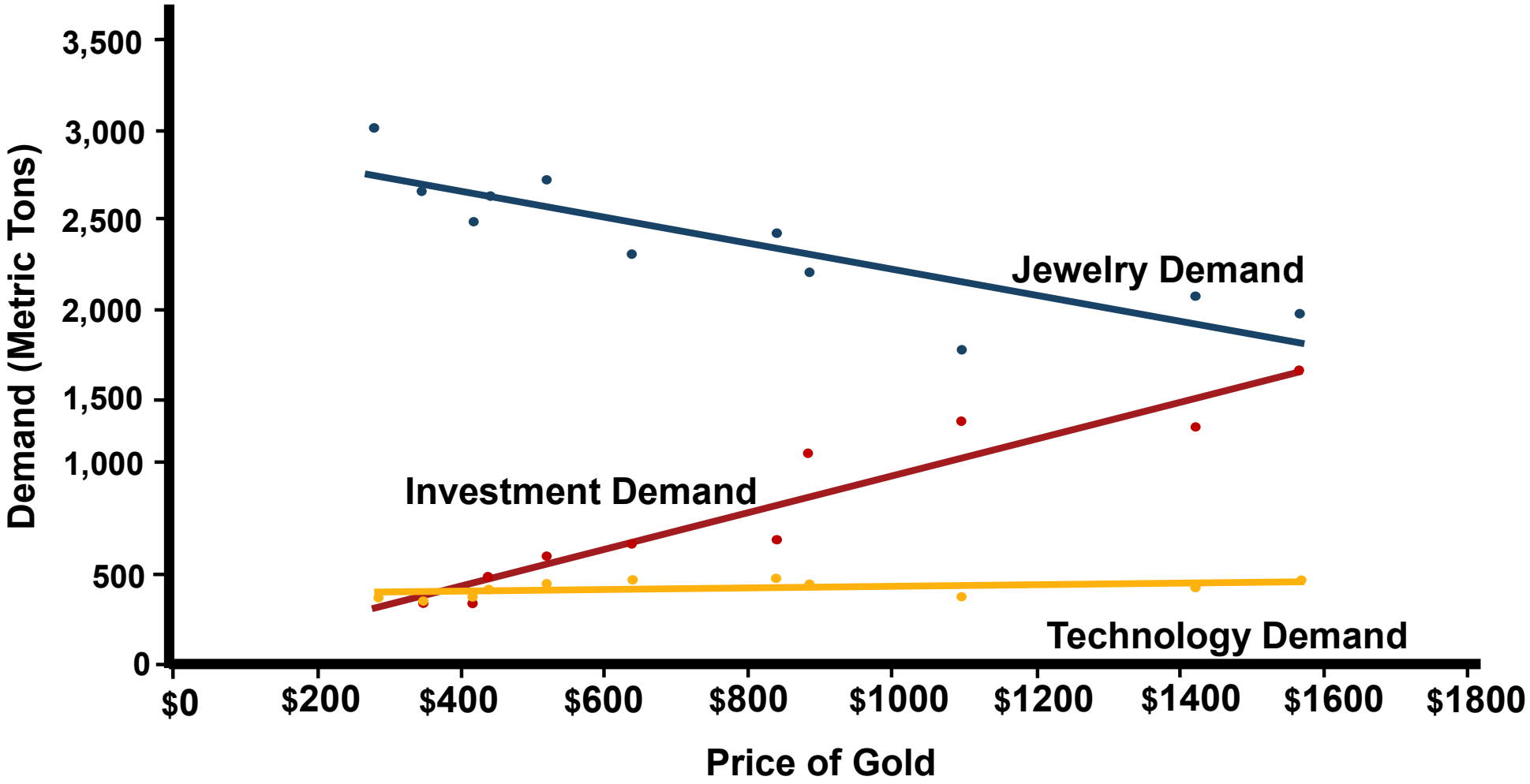




- Estimating the demand curve:
  - Econometric techniques
  - Focus group
  - Market experiment
    - Laboratory
    - Select markets

2.Demand and Supply

- Example 1: Demand curve for gold



- Example 2: Harley-Davidson's complaint about Japanese brands
  - Regression analysis:
    - Left-hand side variable: Harley-Davidson's revenue
    - Right-hand side variable:
      - Manufacturing employment: 0.73
      - Japanese manufacturers' sale: 0.22



- Example 3: Federal Reserve's auto sales in the US

Variable	Regression coefficient	t statistic
Constant	-22,302	-4.5
Disposable income	12.9	6.6
Prime rate	-97.8	-3.2
Inventory-sales ratio	-19.9	-6.1
Auto price	230	5.0
Nonauto price	6.0	2.1

- Example 4: Four scenarios for L'eggs Pantyhose
  - $4 discount for a pack of two
  - $2.5 discount for a pack of two
  - $2 discount for a pair
  - $2.5 rebate for a pair

2.Demand and Supply

- $4 discount for a pack of two: Syracuse, NY
- $2.5 discount for a pack of two: Columbus, OH
- $2 discount for a pair: Denver, CO
- $2.5 rebate for a pair: Cincinnati, OH
- No discount: Boise, ID

2.Demand and Supply

- Factors affecting supply of good X:
  - Price of X
    - The higher/lower the price of X, the higher/lower its supply (direct relationship)
  - Price of inputs (Cost of product)
    - The higher/lower the cost of product, the lower/higher the supply (inverse relationship)
  - Taxes
    - The higher/lower the tax rate, the lower/higher the supply

2.Demand and Supply

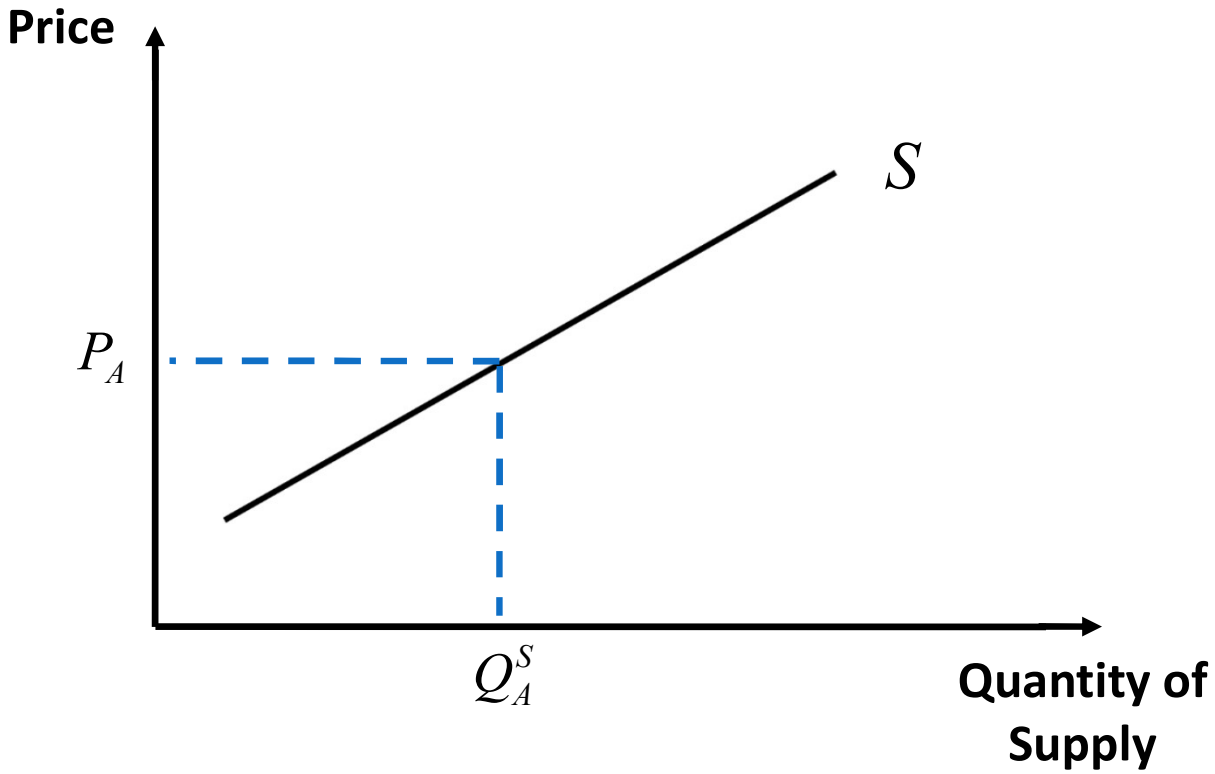
- Supply function:

$$Q_x^S = f(P_x, C, T, \ldots\ldots)$$

- Q_x^S : Quantity of Supply for x
- P_x : Price for x
- C : Cost of Product
- T : Taxes

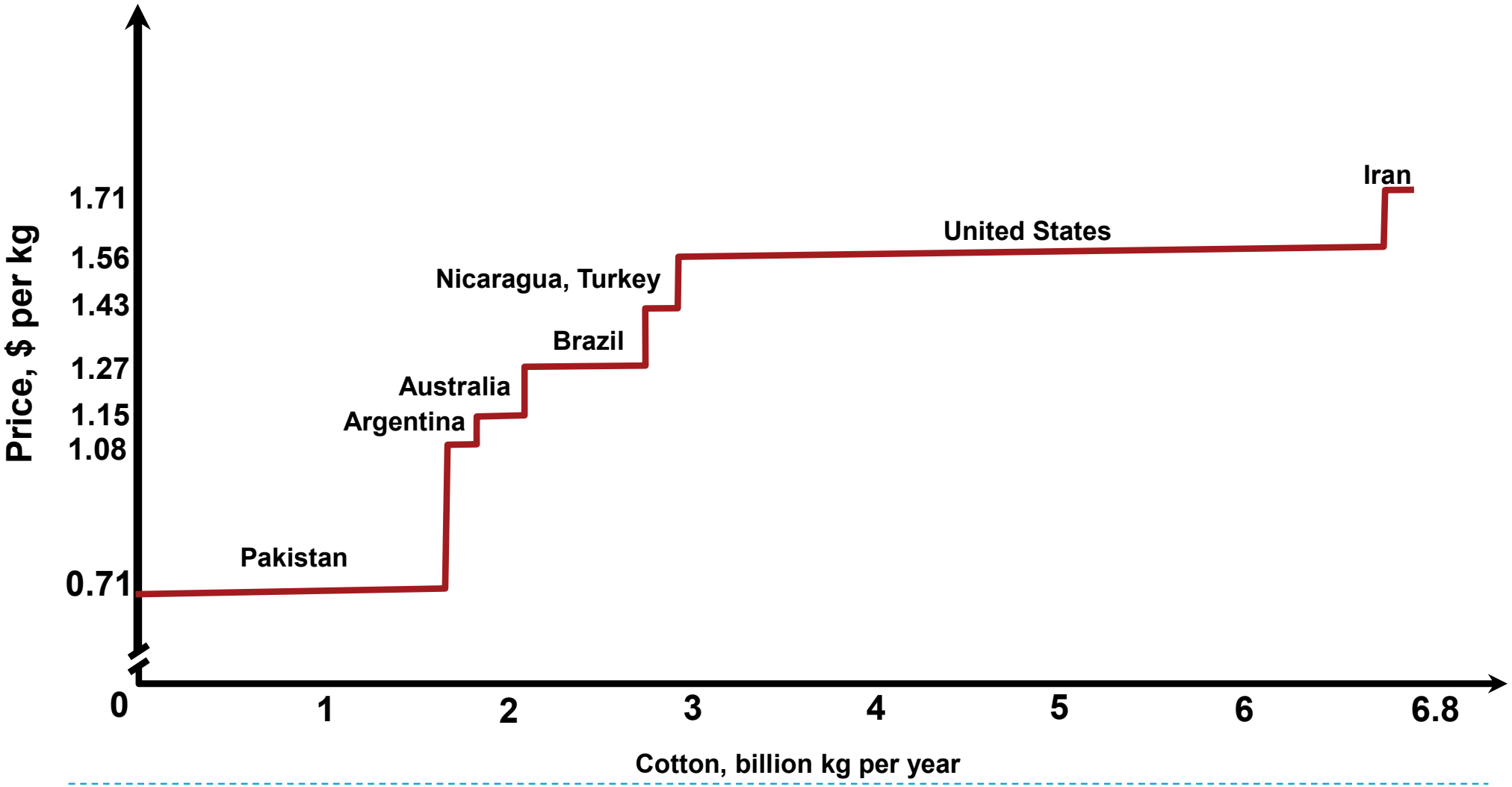
2.Demand and Supply

- Supply Curve: The relationship between price and quantity of supply



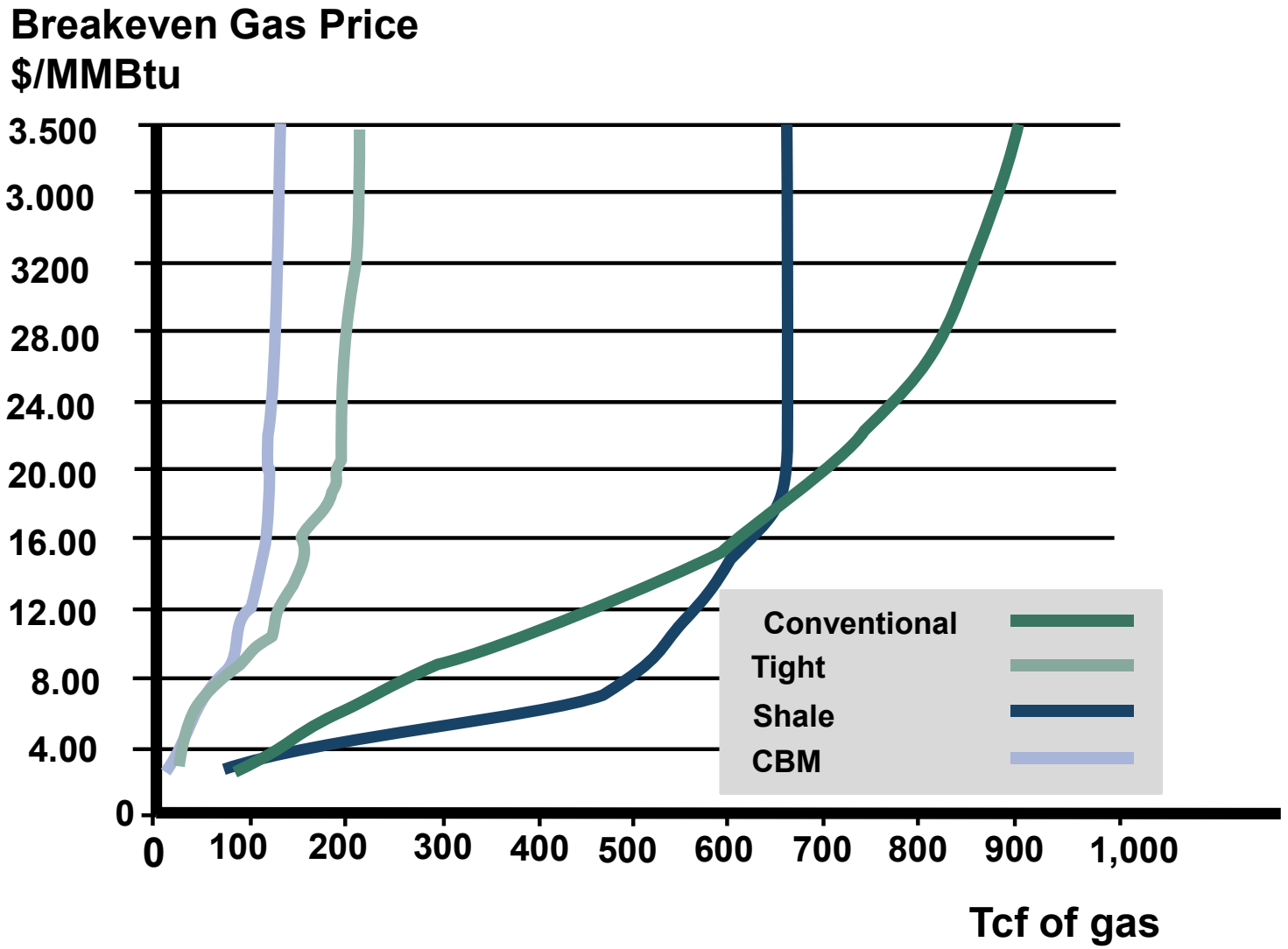
2.Demand and Supply

- Example 1: Global supply curve for cotton:



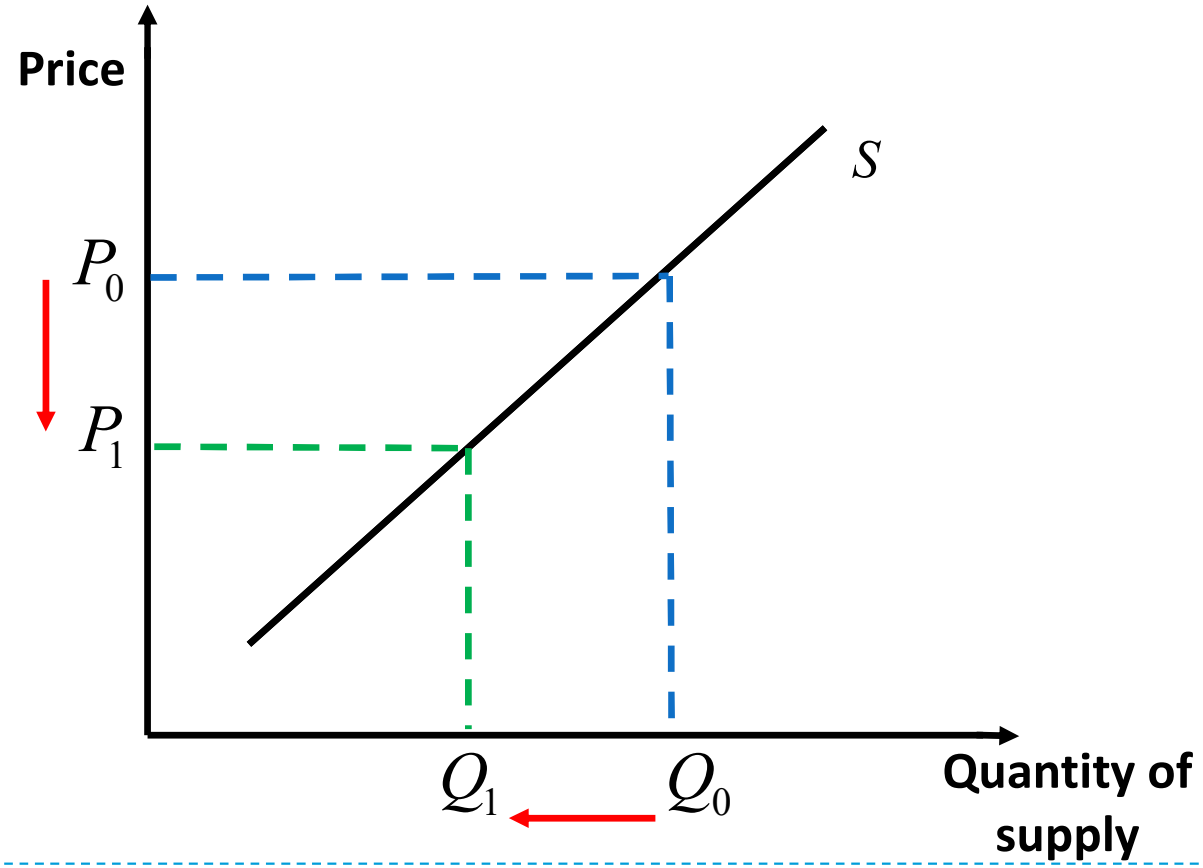
2.Demand and Supply

- Example 2: Global supply curve for gas:



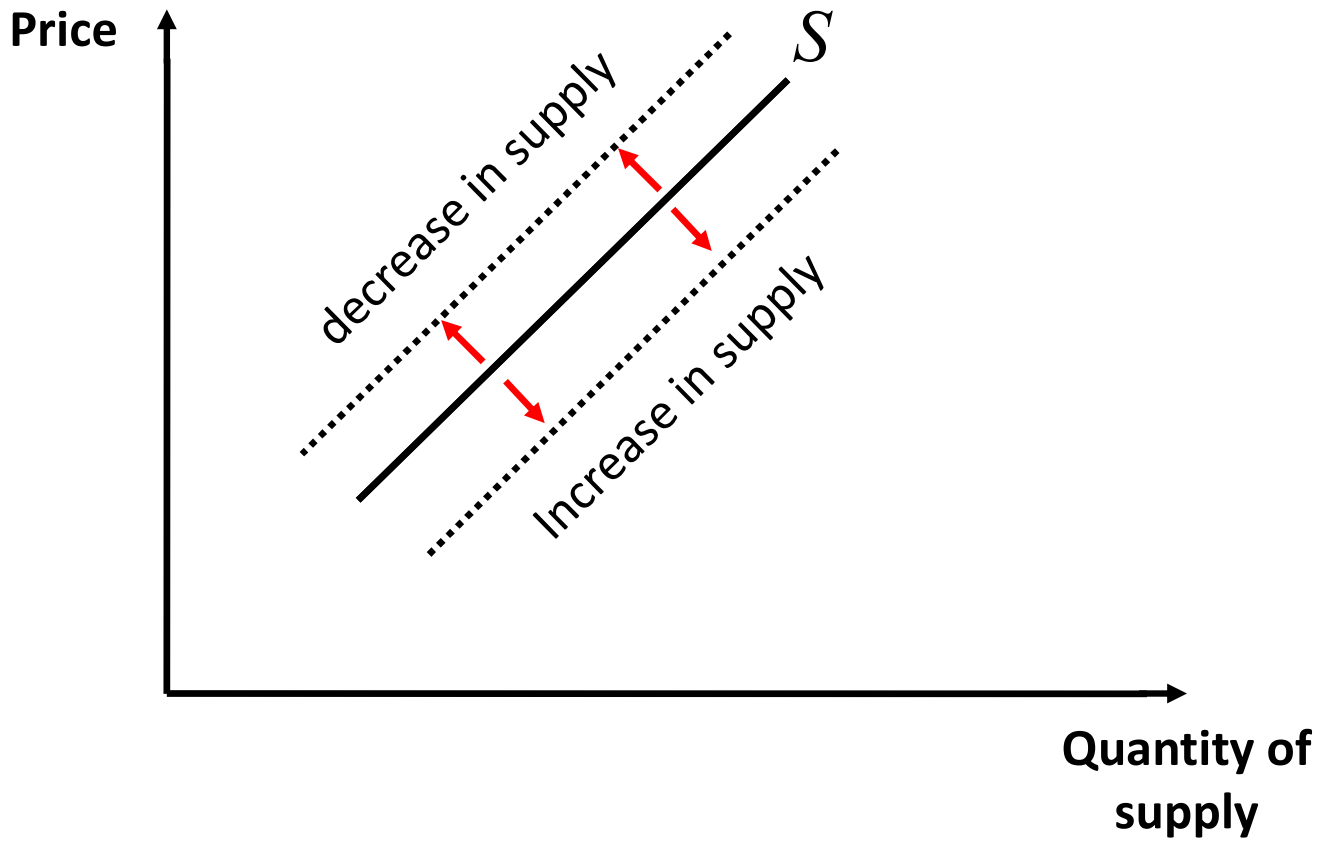
- Two types of changes in supply curve:

1.Change along the supply curve: when price changes holding other factors constant.



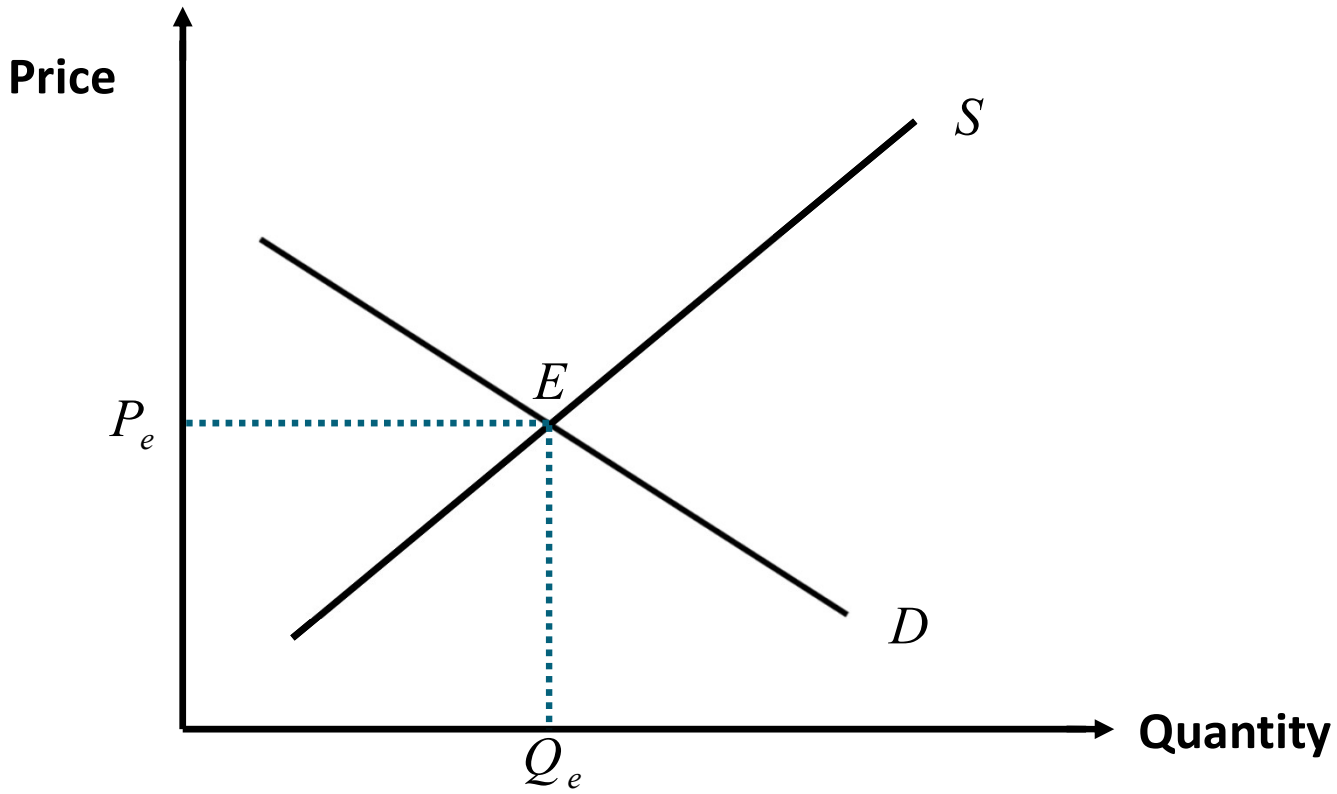
2.Demand and Supply

2.Shift in the supply curve : when other factors change



2.Demand and Supply

- Equilibrium: The point that supply and demand curves intersect



3. Elasticity

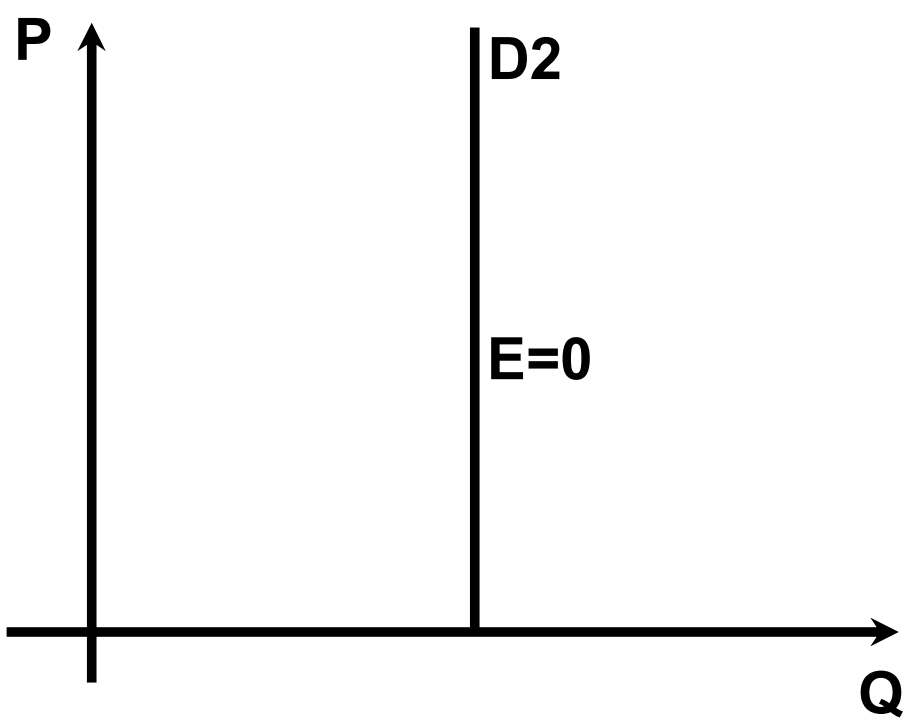
3.Elasticity

- Price elasticity of Demand: a measure of the responsiveness of the quantity demanded to a change in price

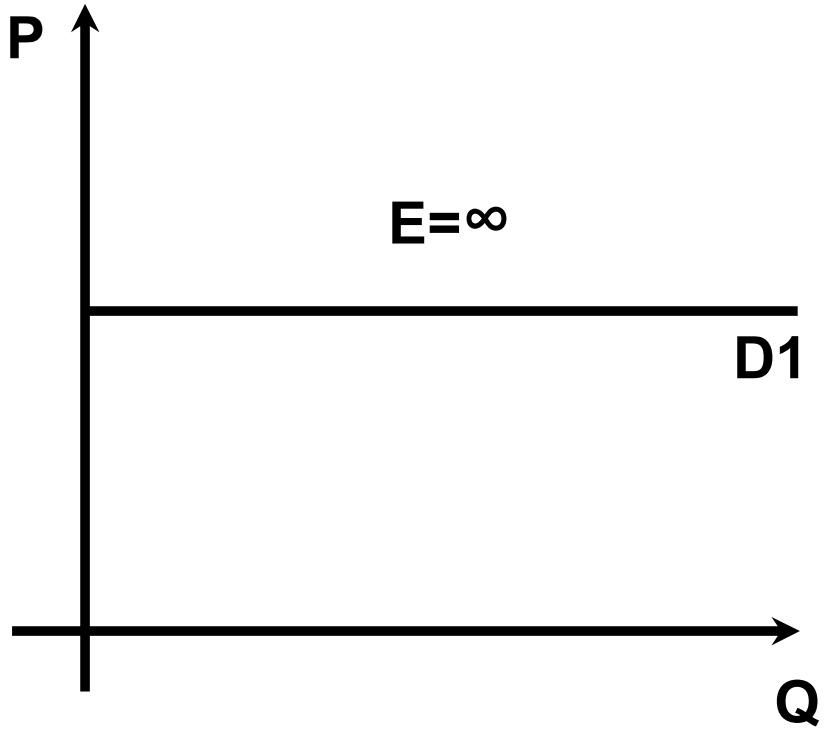
$$E_{Q_x^D, P_x} = \frac{\textbf{\textit{Percentage change in quantity demand}}}{\textbf{\textit{Percentage change in price}}}$$

$$E_{Q_x^D, P_x} = \frac{\%\Delta Q_x^D}{\%\Delta P_x} = \frac{Q_2^D - Q_1^D}{P_2 - P_1} \times \frac{P_2 + P_1}{Q_2^D + Q_1^D}$$

3.Elasticity



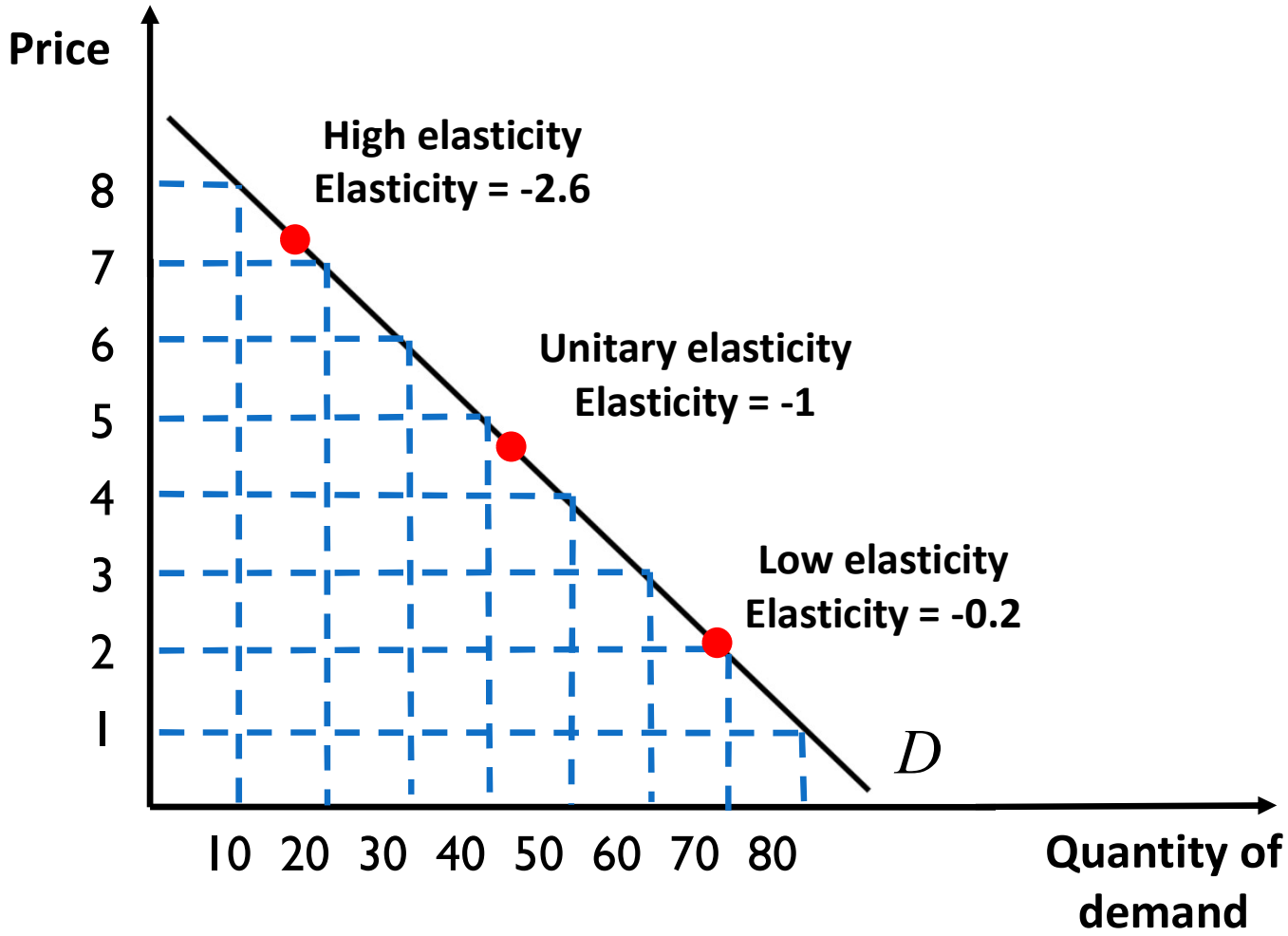
a) Perfectly inelastic demand



b) Perfectly elastic demand

3.Elasticity

- Important: Elasticity is not the slope of the demand curve



3.Elasticity

- Factors affecting demand elasticity:
  - Availability of substitutes
  - Portion of income spent on a good
  - Time

3.Elasticity

- Income elasticity of demand: the sensitivity of quantity demanded to change in income
  - Normal goods : Positive income elasticity
  - inferior goods : Negative income elasticity

$$E_{Q_x^D, I_x} = \frac{\textbf{\textit{Percentage change in quantity demand}}}{\textbf{\textit{Percentage change in income}}}$$

$$E_{Q_x^D, I_x} = \frac{\%\Delta Q_x^D}{\%\Delta I_x} = \frac{Q_2^D - Q_1^D}{I_2 - I_1} \times \frac{I_2 + I_1}{Q_2^D + Q_1^D}$$

3.Elasticity

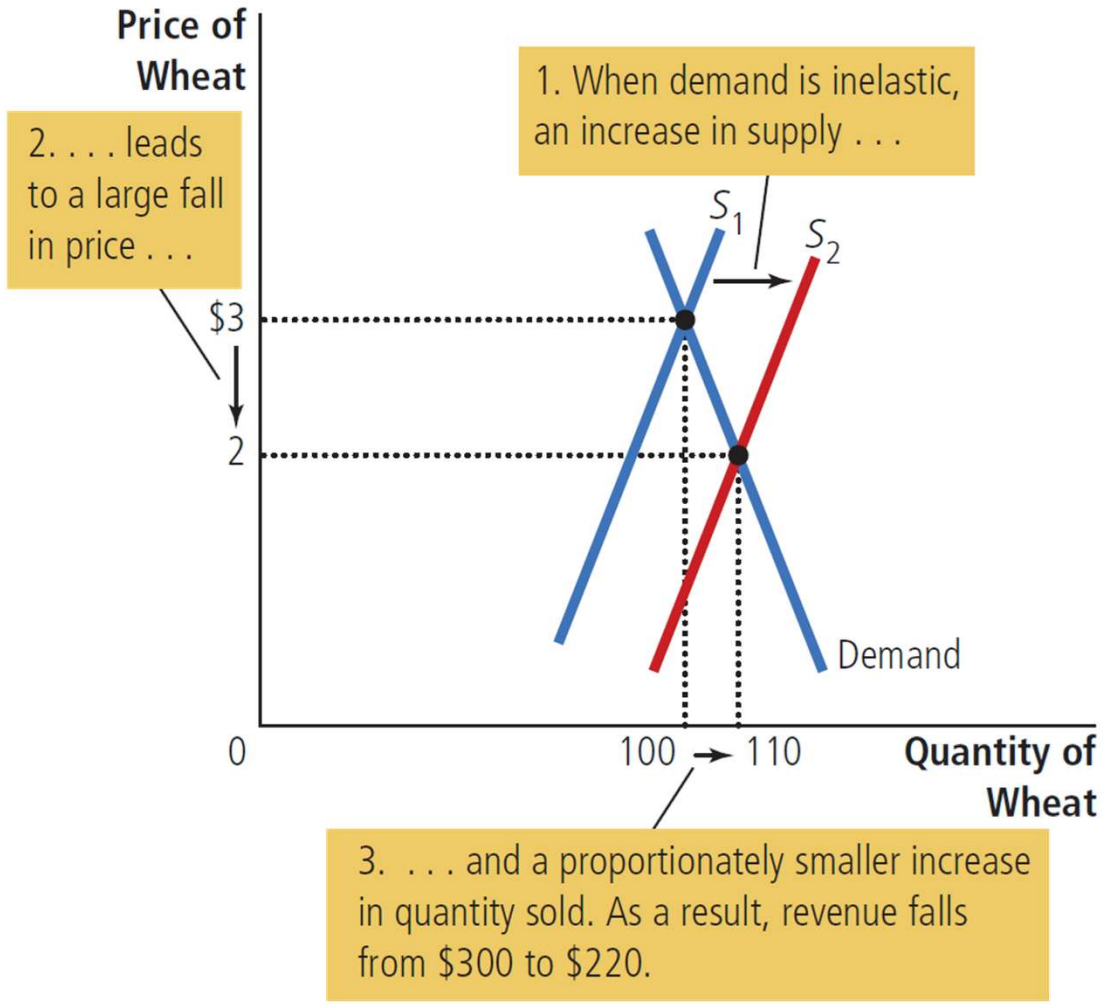
- Cross price elasticity of demand: the ratio of the percentage change in the quantity demanded for a good to the percentage change in the price of a related good
  - Substitutes : positive cross elasticity
  - Complements : negative cross elasticity

$$E_{Q_x^D, P_y} = \frac{\textit{Percentage change in quantity demand for good x}}{\textit{Percentage change in price for good y}}$$

$$E_{Q_x^D, P_y} = \frac{\%\Delta Q_x^D}{\%\Delta P_y} = \frac{Q_2^x - Q_1^x}{P_2^y - P_1^y} \times \frac{P_2^y + P_1^y}{Q_2^x + Q_1^x}$$

3.Elasticity

- Example 1: The effect of productivity improvement on farmers' income



3.Elasticity

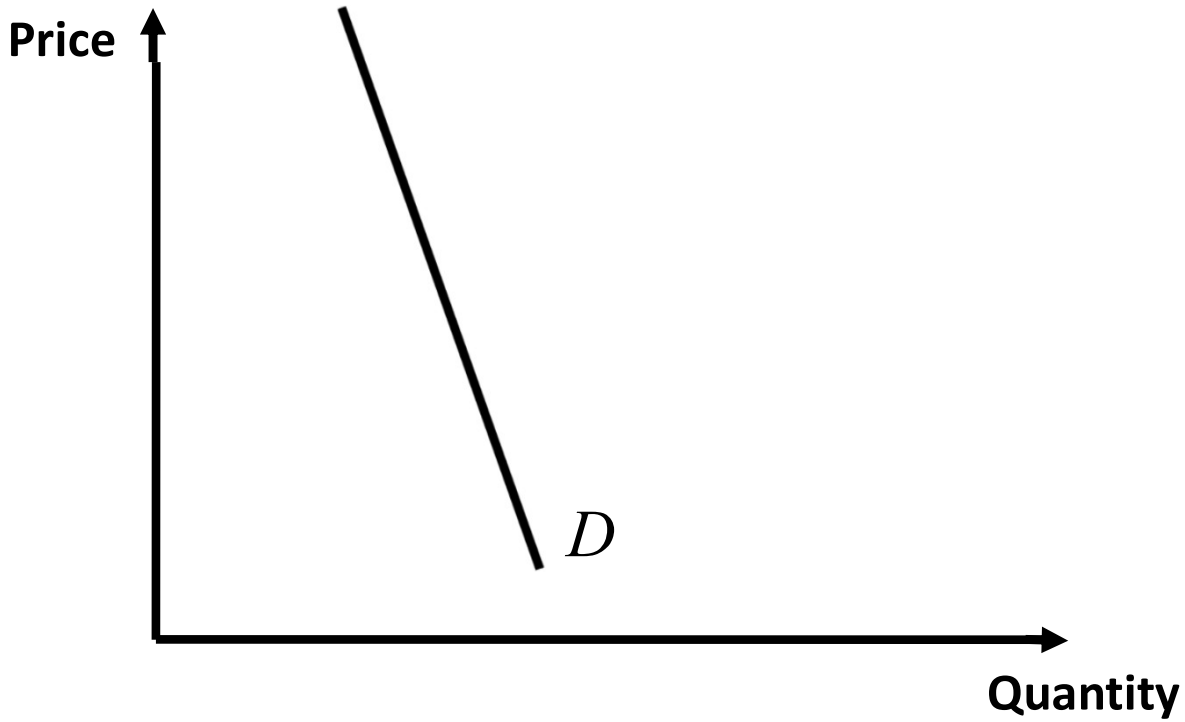
- Question: If improvements in productivity make farmers worse off, why would farmers adopt new technology?
  - Answer: Competition!
    - Each farmer takes the price of wheat as given.
  - In 1950, 17% of labor force worked on farms, whereas today, only 3% work on farms.
  - Today, US farms produce five times as much outputs as they did in 1950.

3.Elasticity

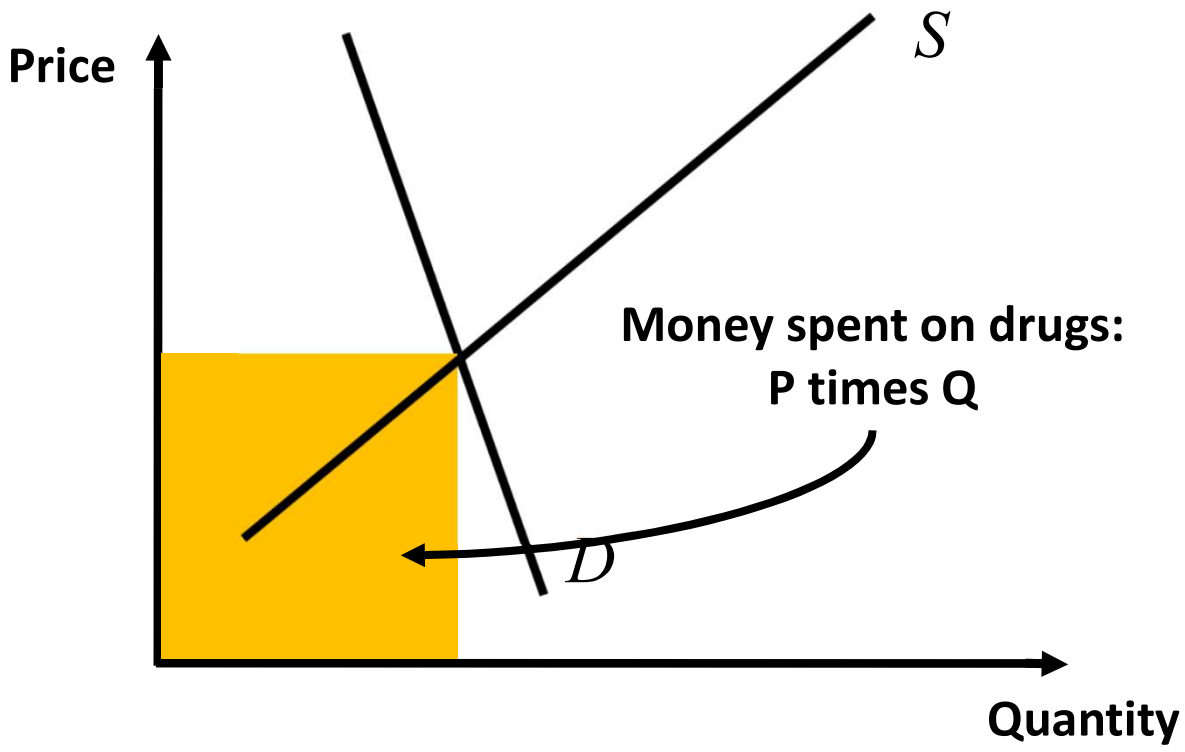
- Example 2: Fighting against drug dealing and the related crimes
- Assumptions:
  - Crimes related to drug dealing are correlated with financial turnover.
  - Government's resources are limited.
- Policy question: Are supply-side or demand-side policies more effective?

3.Elasticity

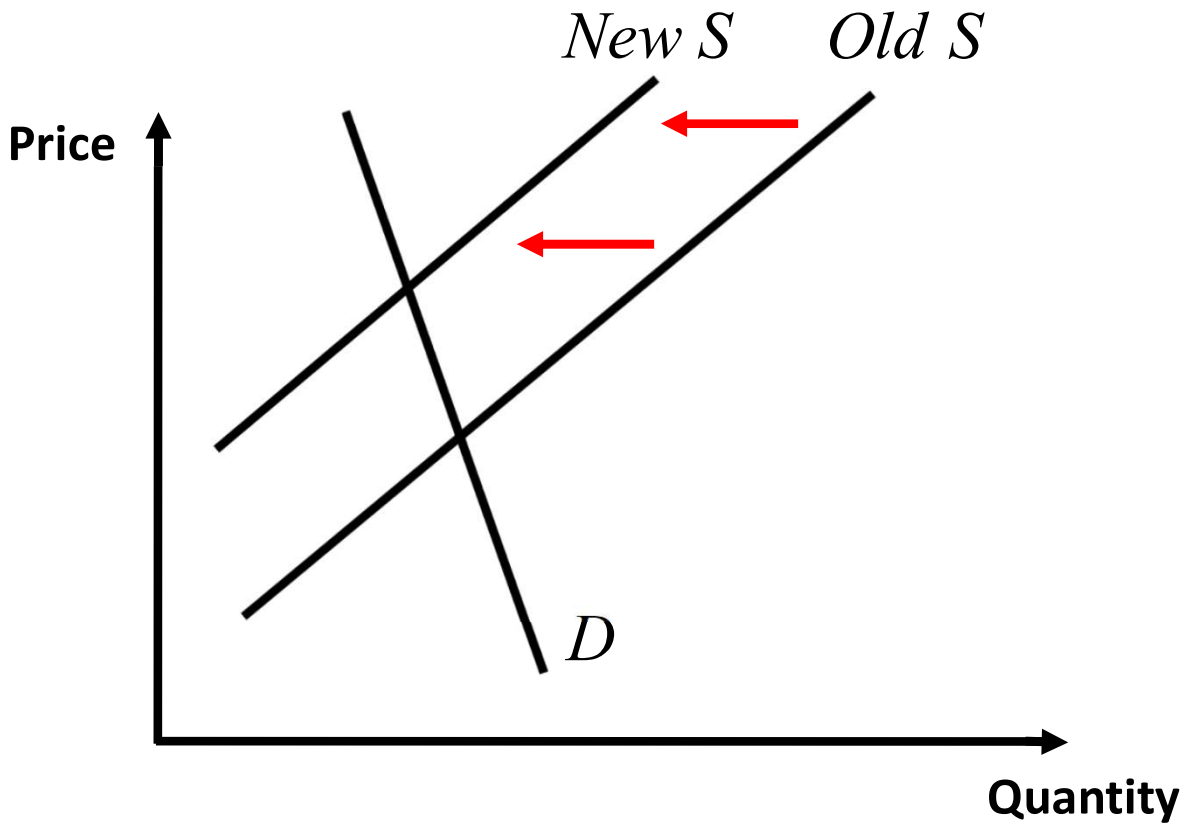
- Demand curve:
  - Demand for drug is likely inelastic



- Supply and demand equilibrium:

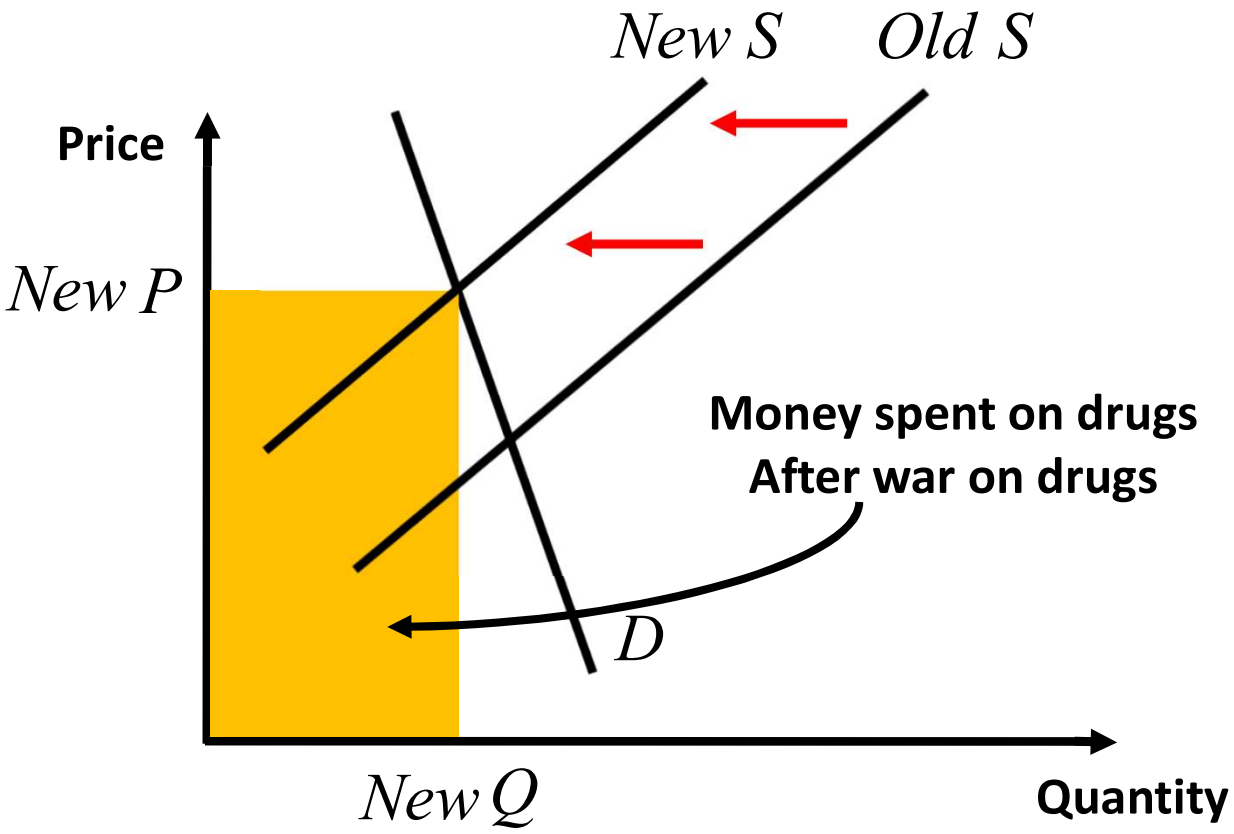


- Supply-side policy: Arresting and prosecuting drug dealers

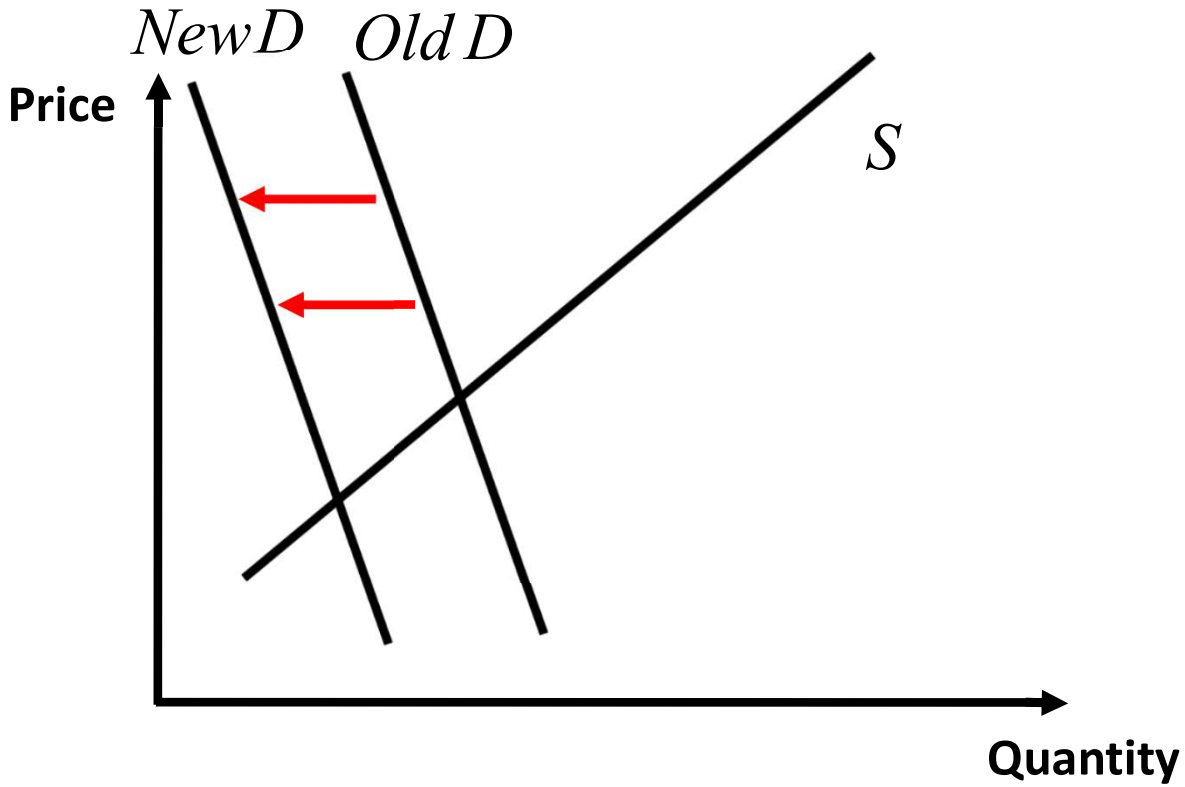


3.Elasticity

- Supply-side policy outcome: Increase in money turnover

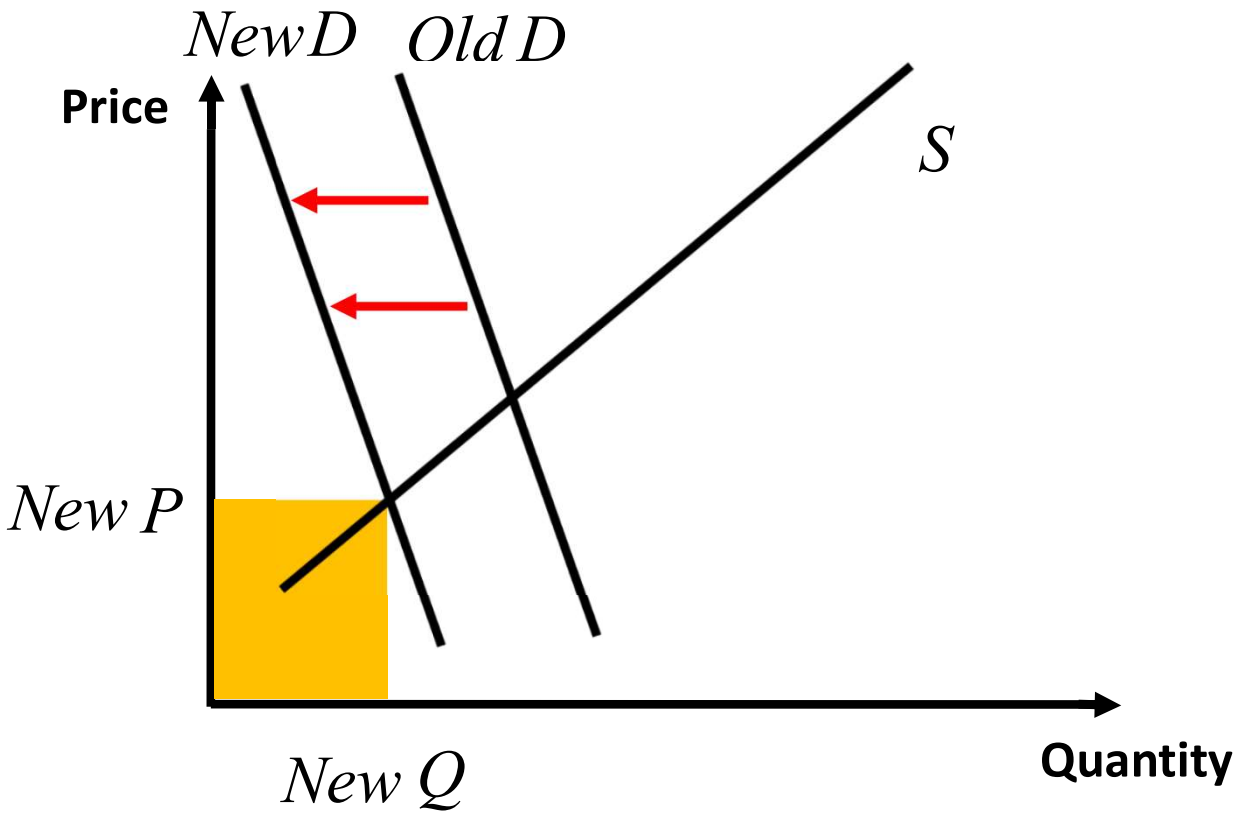


- Demand-side policy: Educating the public about the consequences



3.Elasticity

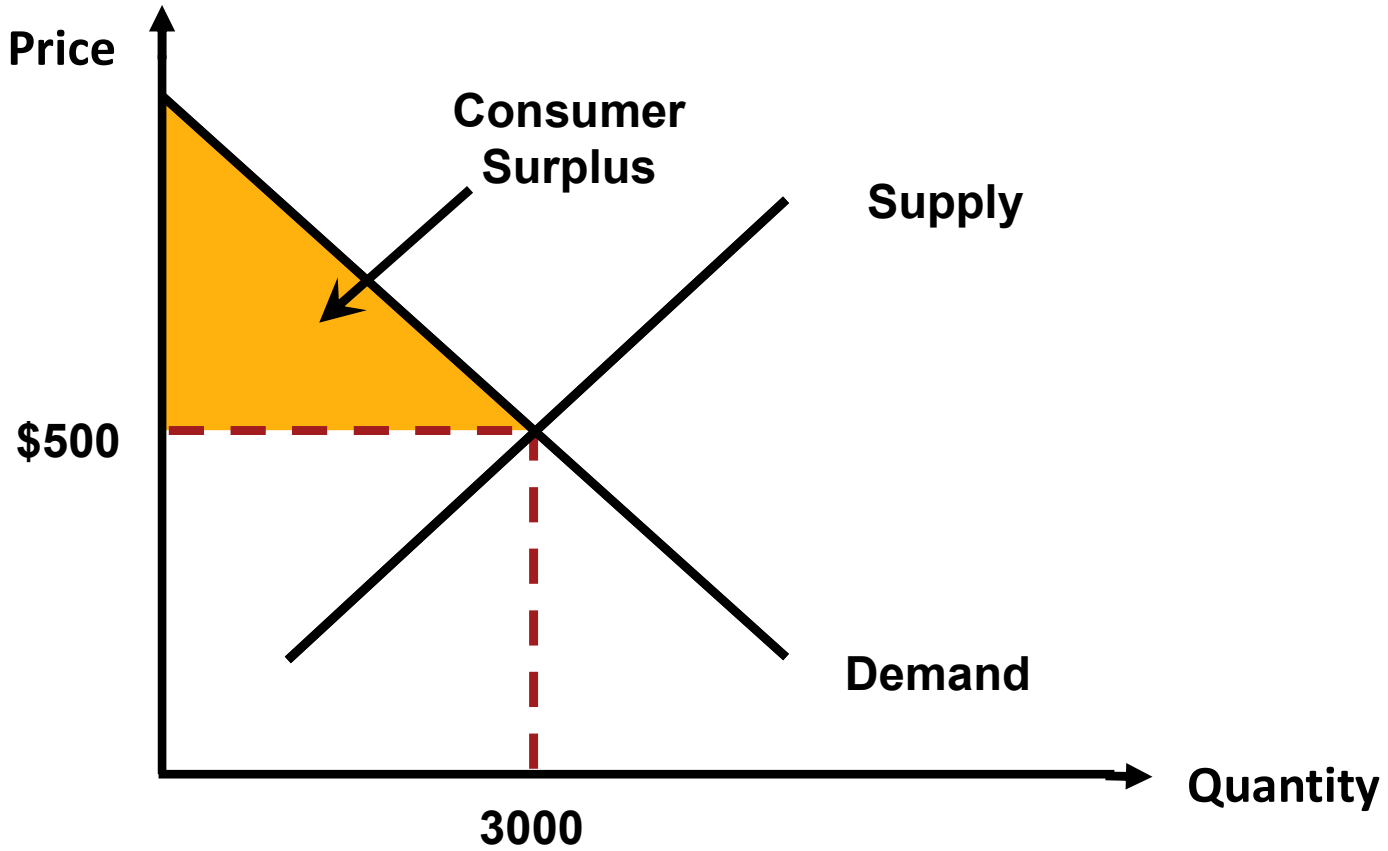
- Demand-side policy outcome: Reduction in money turnover



4.Surplus and deadweight loss

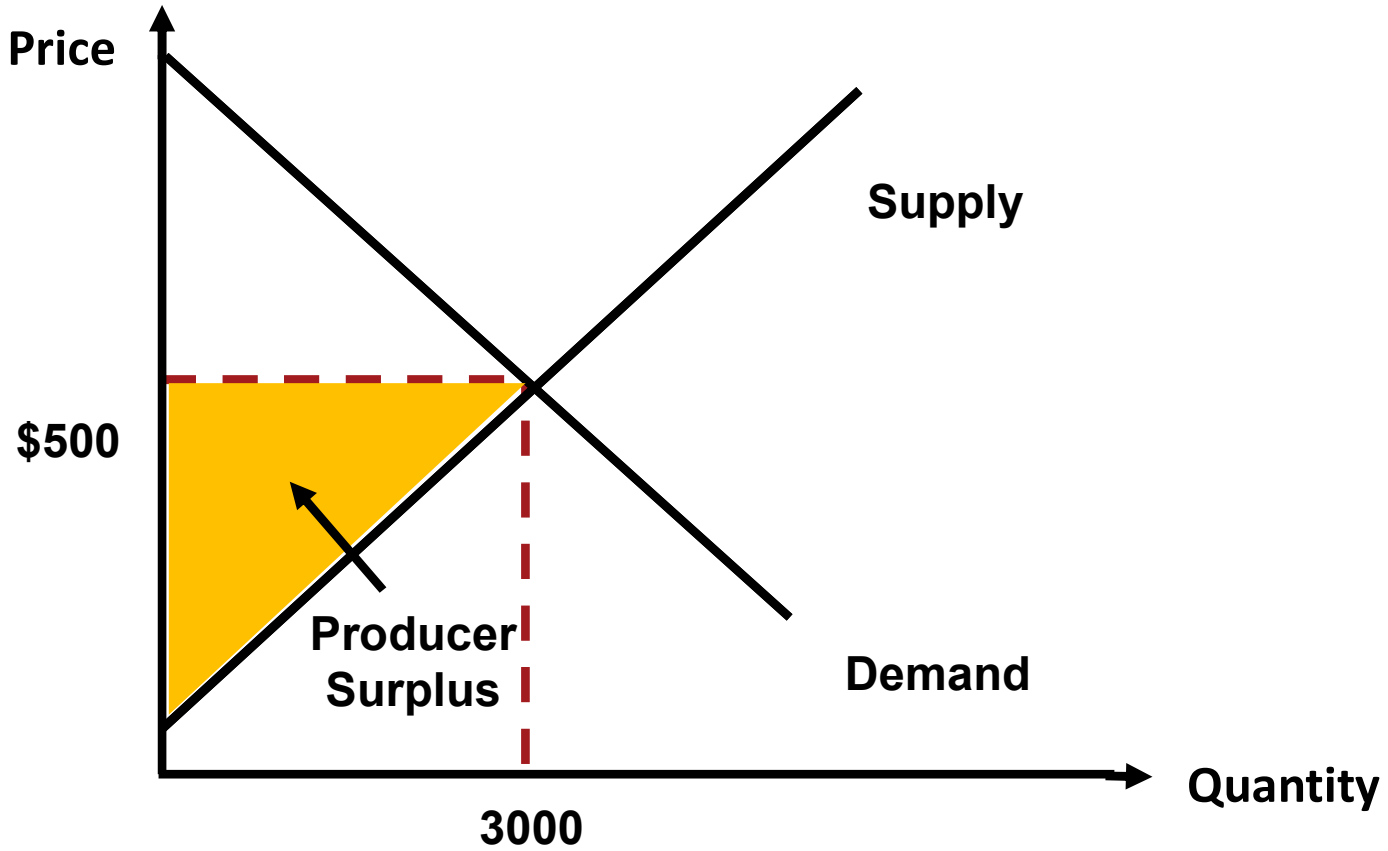
4.Surplus and deadweight loss

- Consumer surplus: Difference between total value for consumers of the units of a good that they buy and the total amount they must pay for those units



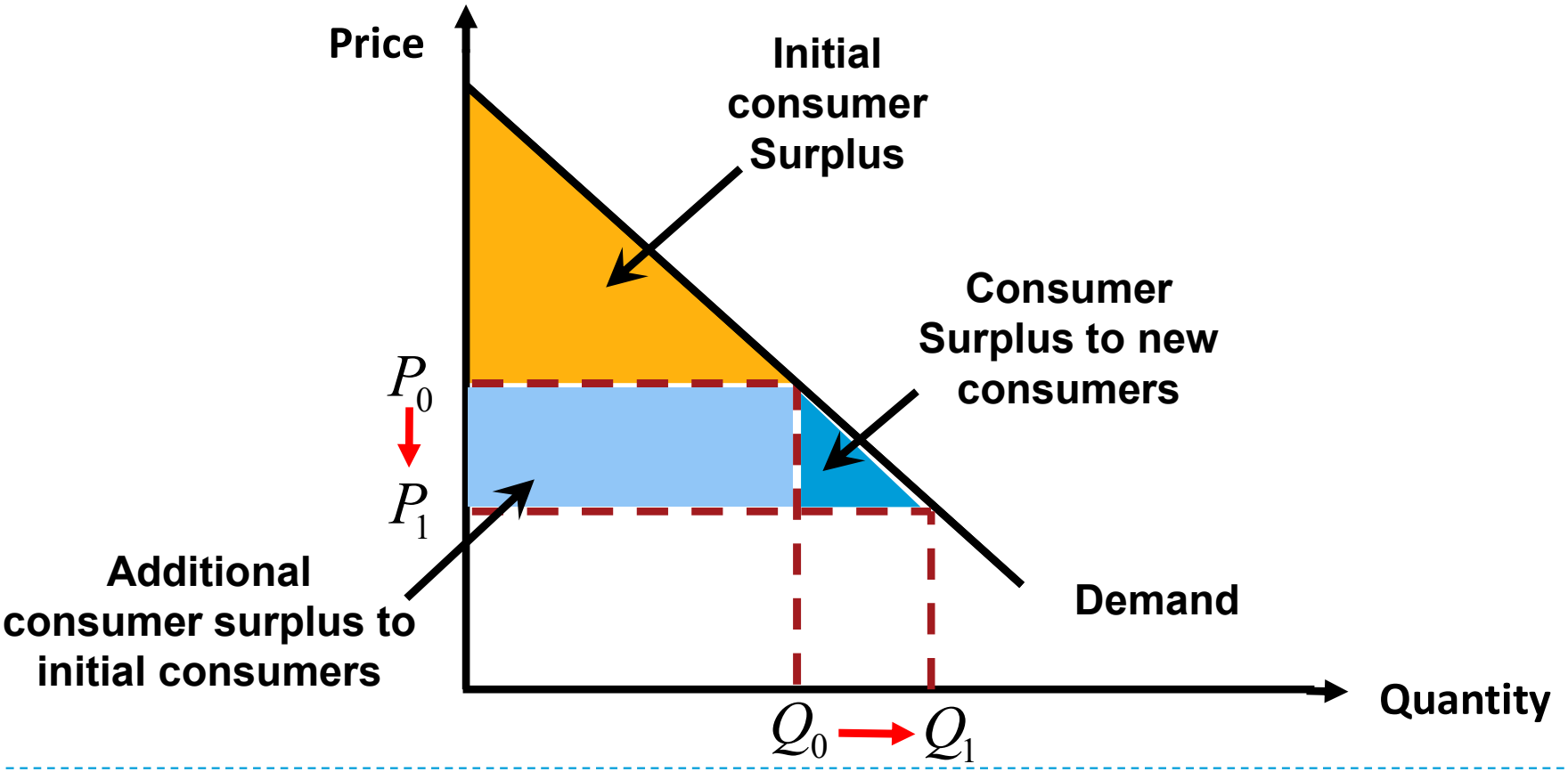
4.Surplus and deadweight loss

- Producer surplus: the excess of the market price above the opportunity cost of production



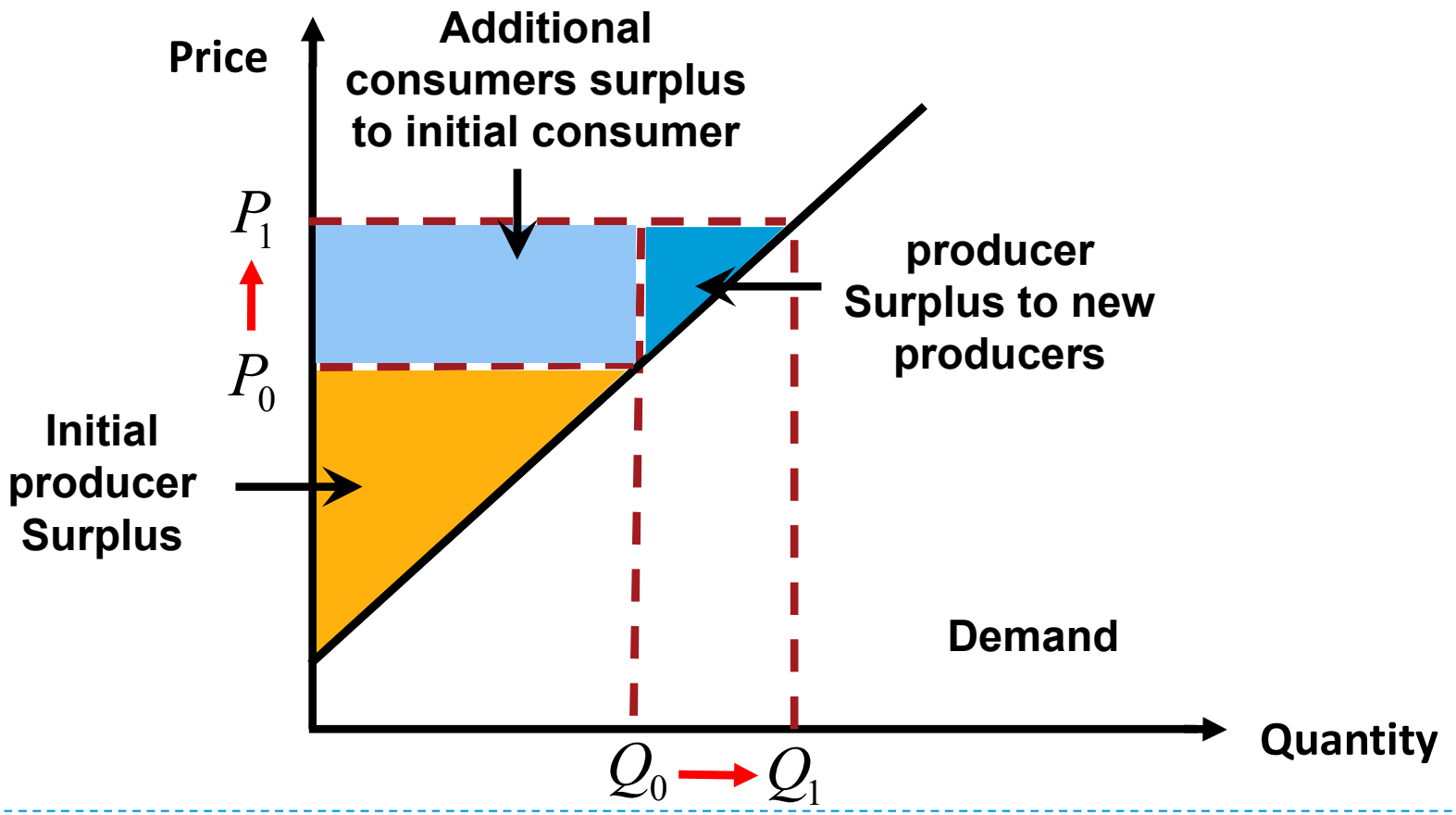
4.Surplus and deadweight loss

- Change in consumer surplus due to price change:



4.Surplus and deadweight loss

- Change in producer surplus due to price change:



4.Surplus and deadweight loss

- Example: Introducing Apple Cinnamon Cheerios
  - $60M surplus per year



4.Surplus and deadweight loss

- Examples of price discrimination by producers to maximize producer surplus:
  - Airlines
  - Cinemas and restaurants
  - Disneyland
  - Black Friday
  - Accent and taxi drivers!
  - ….

4.Surplus and deadweight loss

- Why do sellers discriminate when setting prices?
  - Improve margins
  - Incentivize buyers to buy differently or more
  - Sell to new customer segments with different WTP.
  - Cost-based price discrimination
- Why do buyer accept price discrimination?
  - Price is higher: Price is less important than other factors (e.g. convenience), resale is not possible
  - Price is lower: Sweet deal!

A Segway to Pricing

- Three approaches to pricing:
  - Cost-plus pricing
    - Full cost per unit + Markup = Selling price
  - Value-based pricing
  - Market-based pricing

A Segway to Pricing

- Cost-plus Pricing is useful when:
  - Sellers have sufficient pricing power
  - Transactions with very custom specifications
  - Price-regulated industries
  - As a starting point for a new product, etc.

A Segway to Pricing

- Cost-plus Pricing:



Advantages

- Guarantees a targeted margin
- Intuitive
- Easy to defend
- Simple to execute

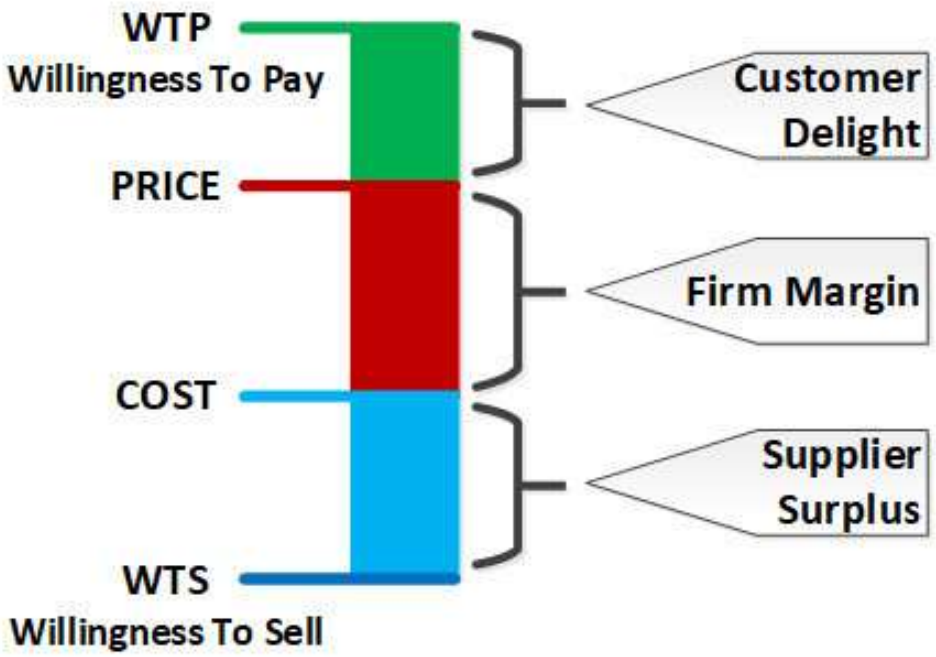


Disadvantages

- Usually sub-optimal pricing
- Ignores opportunity cost
- Uses historical cost instead of current value
- Promotes cost inefficiency

4. Another Look at Consumer and Producer Surplus

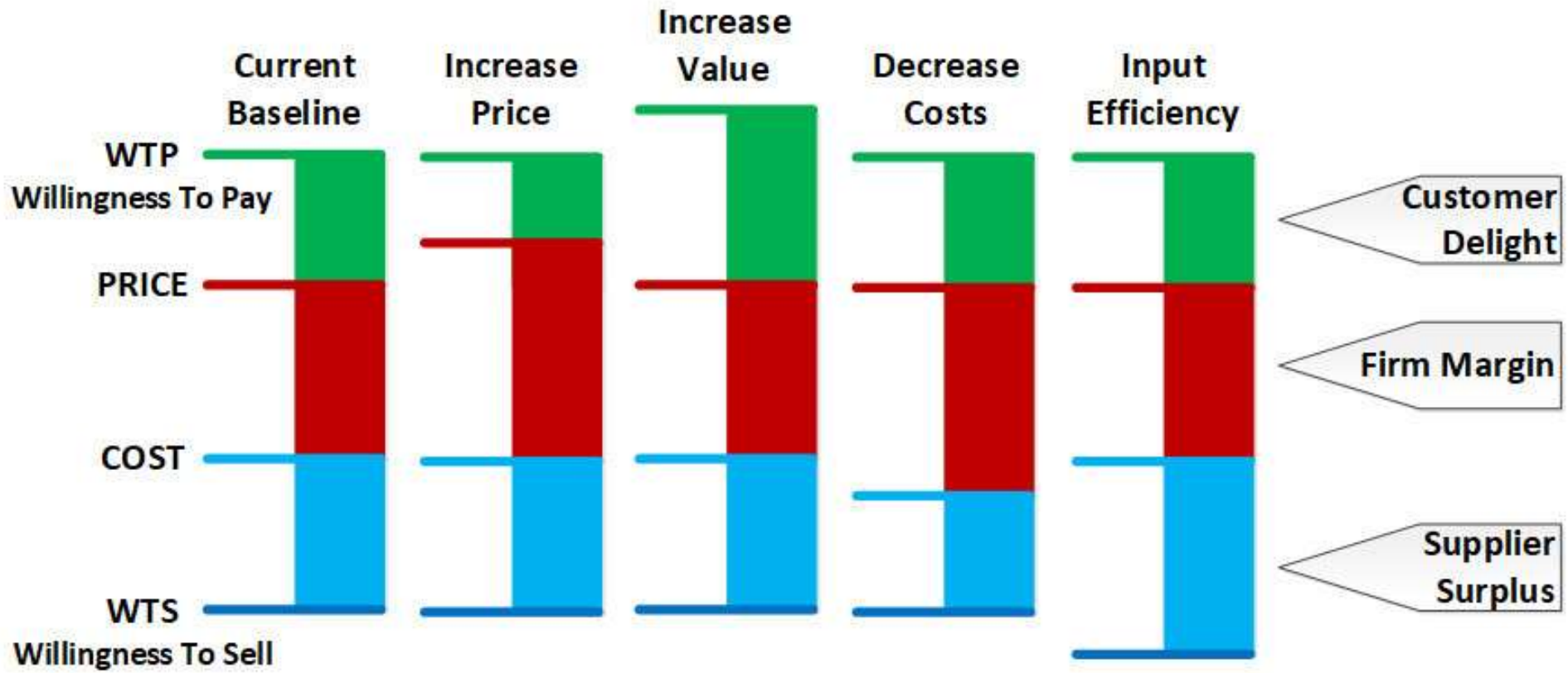
- The Value Stick is a tool that provides insight into where the value is in a product or service.



Source: The Strategic Project Manager

4. Another Look at Consumer and Producer Surplus

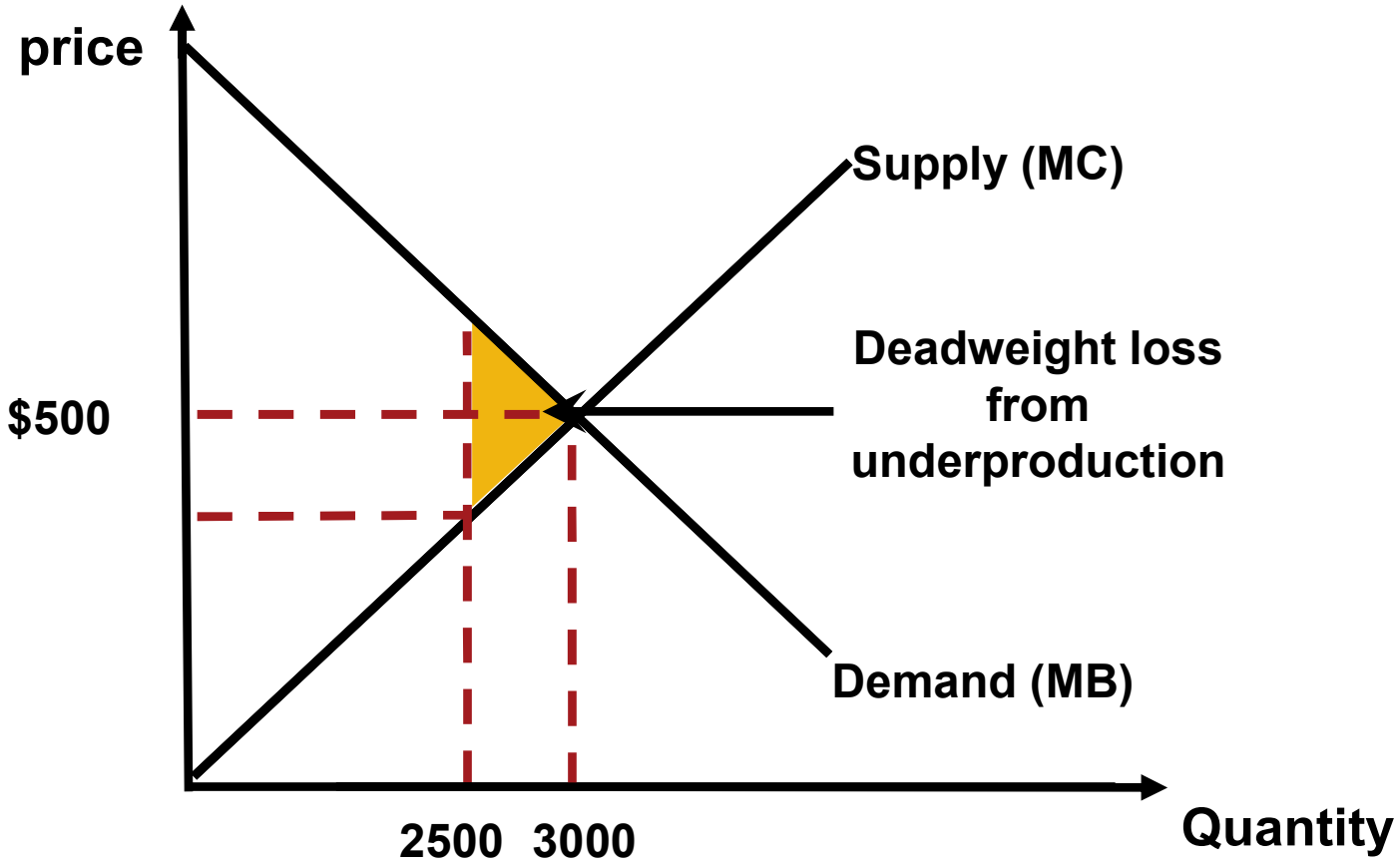
- Value Stick Framework Dynamics in four scenarios:



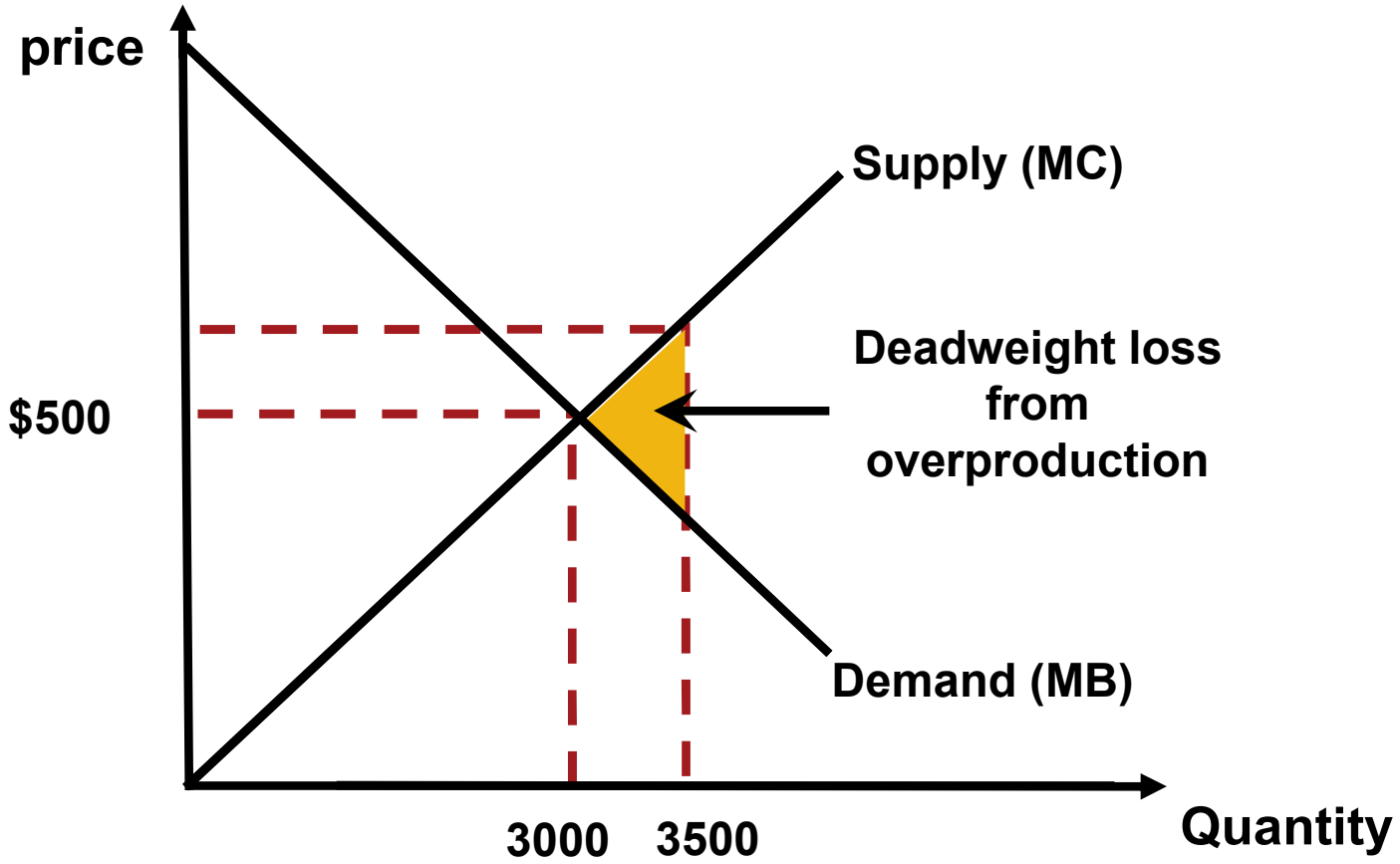
Source: The Strategic Project Manager

4.Surplus and deadweight loss

- Deadweight loss: Reduction in consumer and producer surplus due to underproduction or overproduction



4.Surplus and deadweight loss



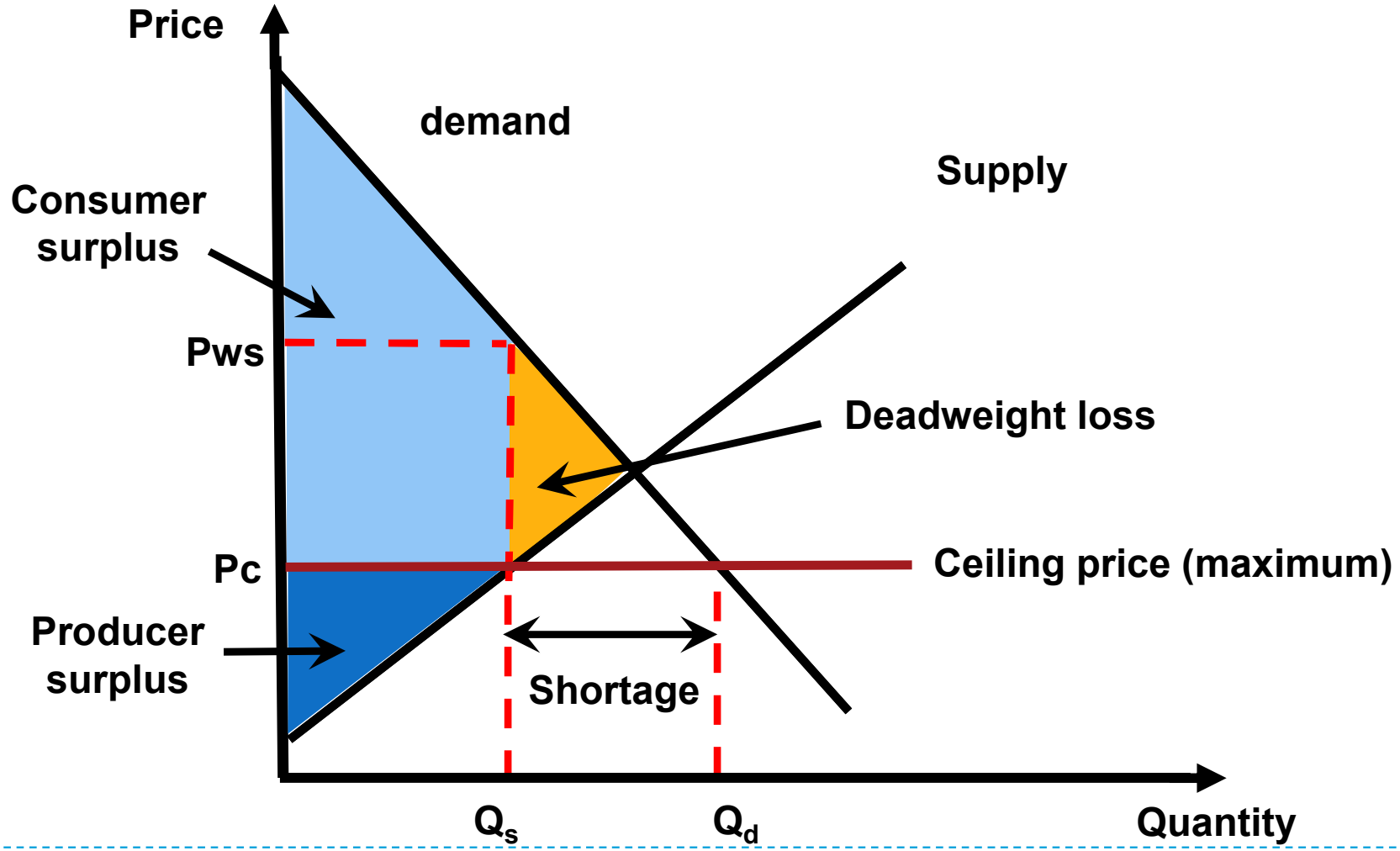
5.Obstacles to efficiency

5. The obstacles to efficiency

- The obstacles to efficiency:
  - Price controls (Price ceiling and Price floor)
  - Taxes and Trade restrictions
  - External Costs
  - External benefits
  - Public goods and common resources
  - High transaction cost

5.The obstacles to efficiency

- Price ceiling: Upper limit on the price



5.The obstacles to efficiency

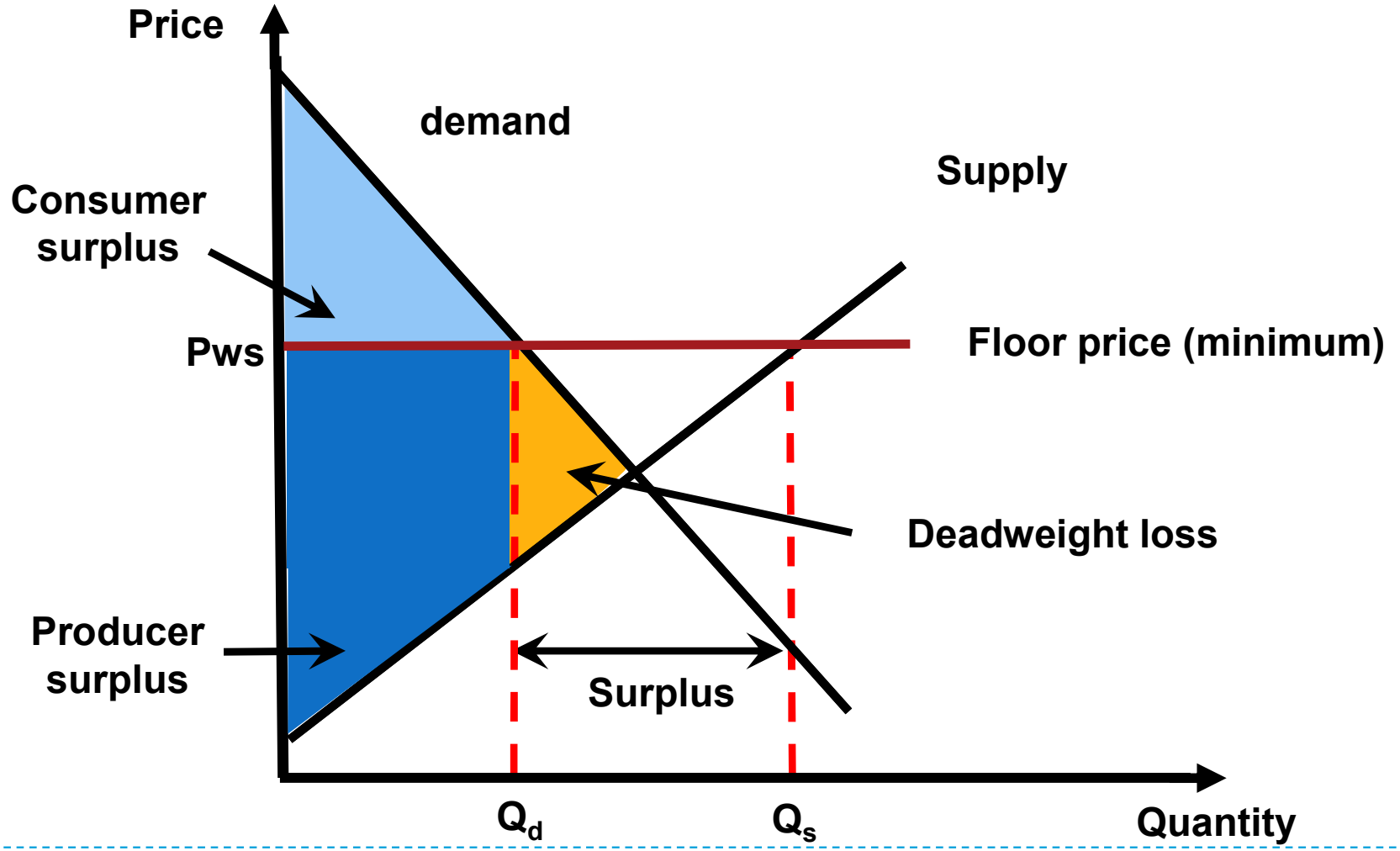
- Examples of price ceilings:
  - Gas prices in the US in the 1970s (and in Iran today!)
  - Rent control
  - ….

5.The obstacles to efficiency

- Long-run impacts of price ceilings:
  - Consumers may have to wait in long lines
  - Suppliers may engage in discrimination
  - Suppliers may take bribes
  - Suppliers may reduce the quality

5.The obstacles to efficiency

- Price floor: minimum price a supplier can offer

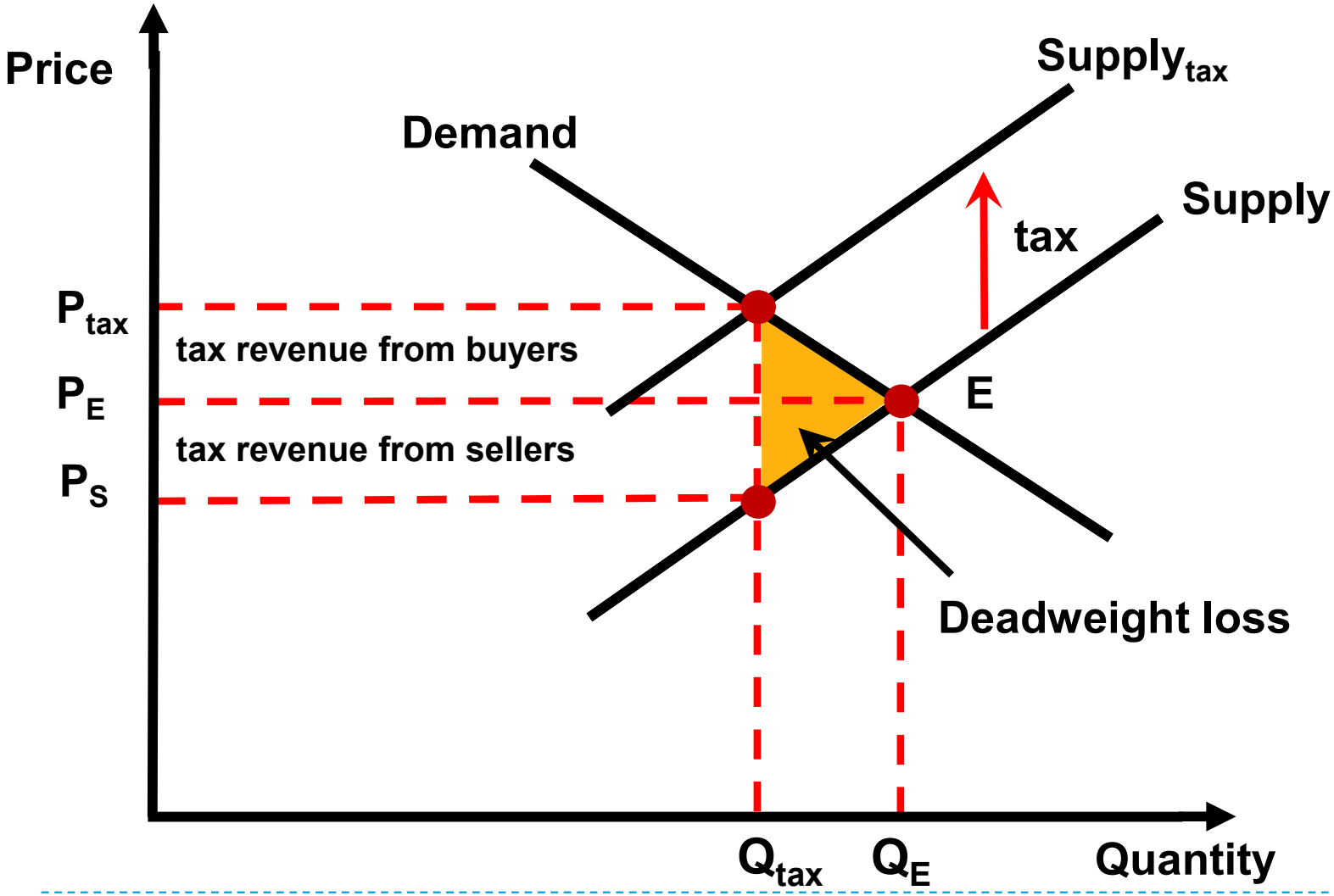


5.The obstacles to efficiency

- Long-run impacts of price floors:
  - Suppliers divert more resources to production but will not be able to sell
  - Consumers will switch to alternatives

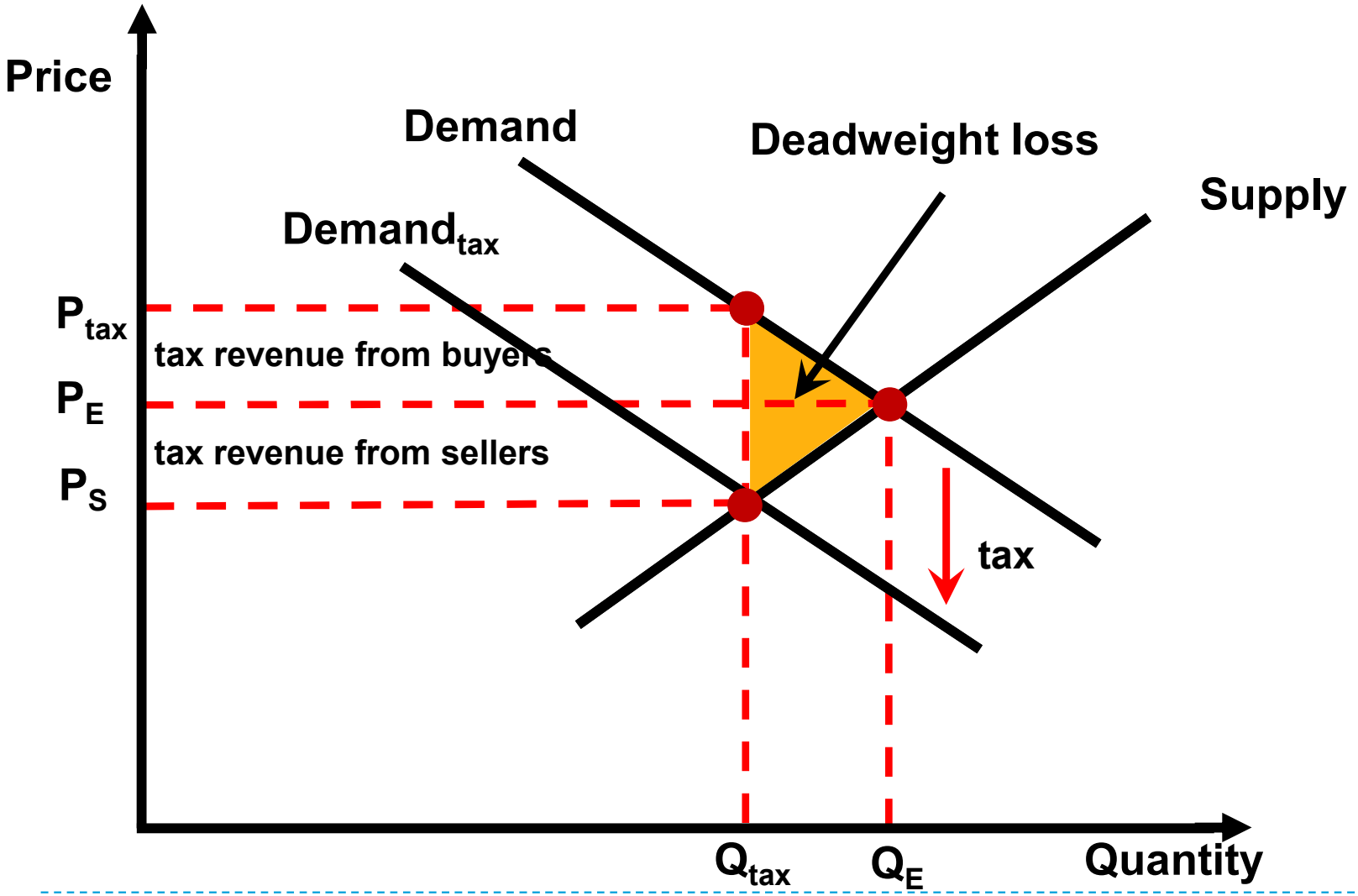
5.The obstacles to efficiency

- Tax on producers:



5.The obstacles to efficiency

- Tax on buyers:



5. The obstacles to efficiency

- Statutory incidence: who is legally responsible for paying the tax

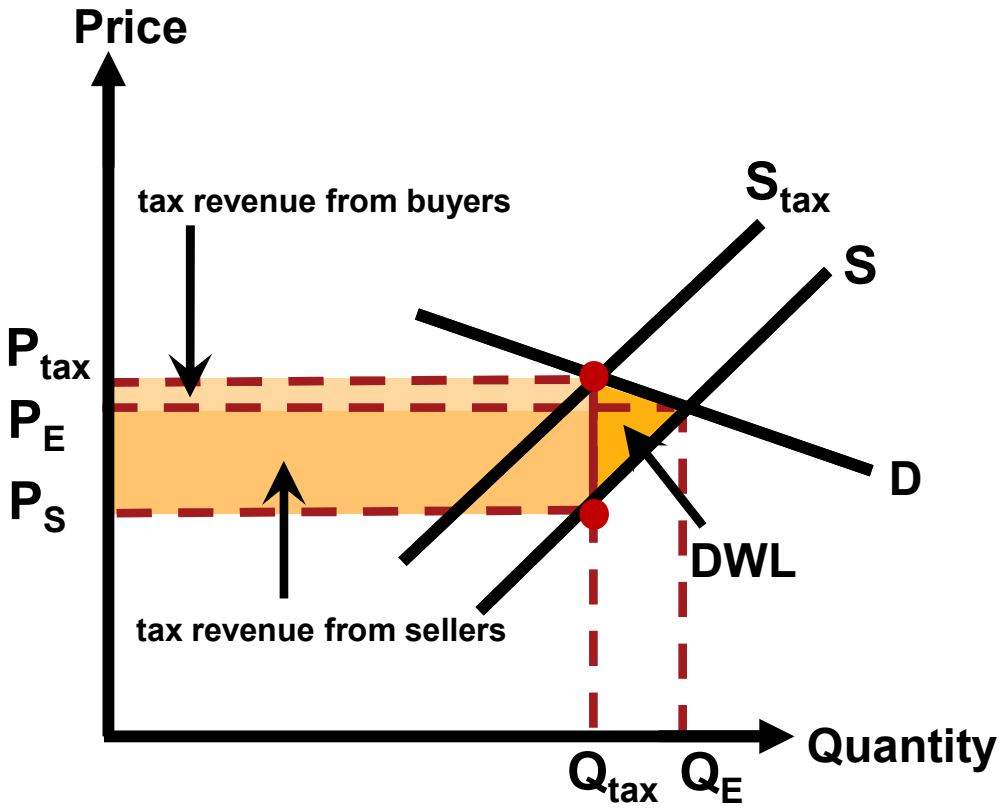
- Actual incidence of tax: who actually bears the cost of tax

- Note: Elasticity of supply and demand determines the actual incidence of tax, not the statutory incidence

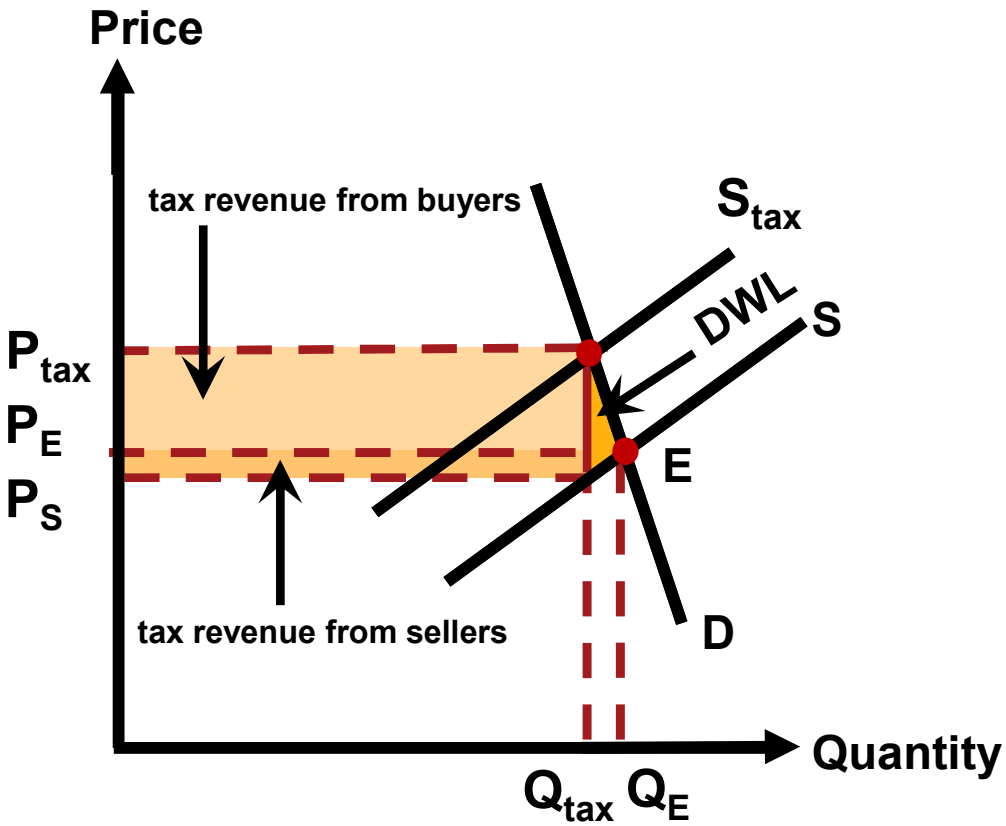
5.The obstacles to efficiency

- If demand is less elastic than supply, consumers will bear a higher burden (Example: cigarette vs. gasoline)

a) Elastic Demand Curve

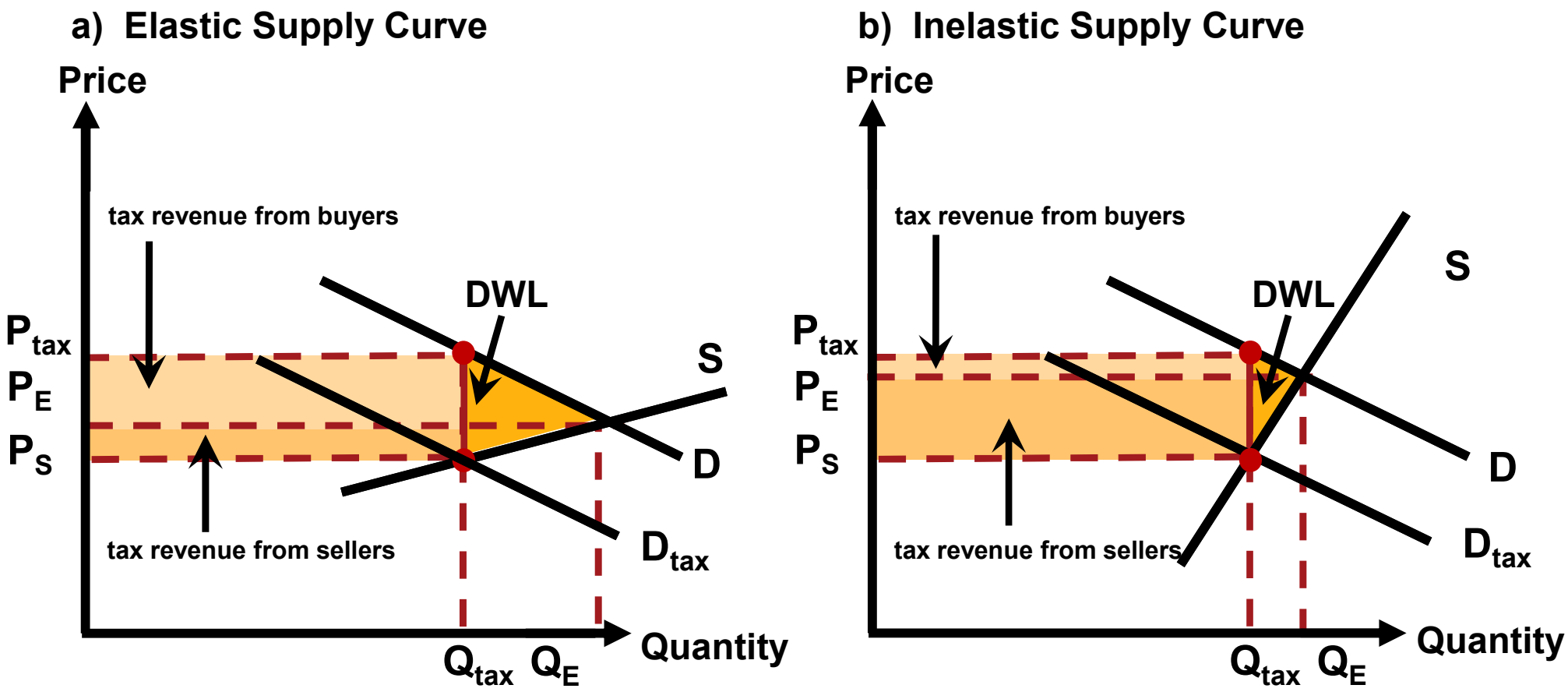


b) Inelastic Demand Curve



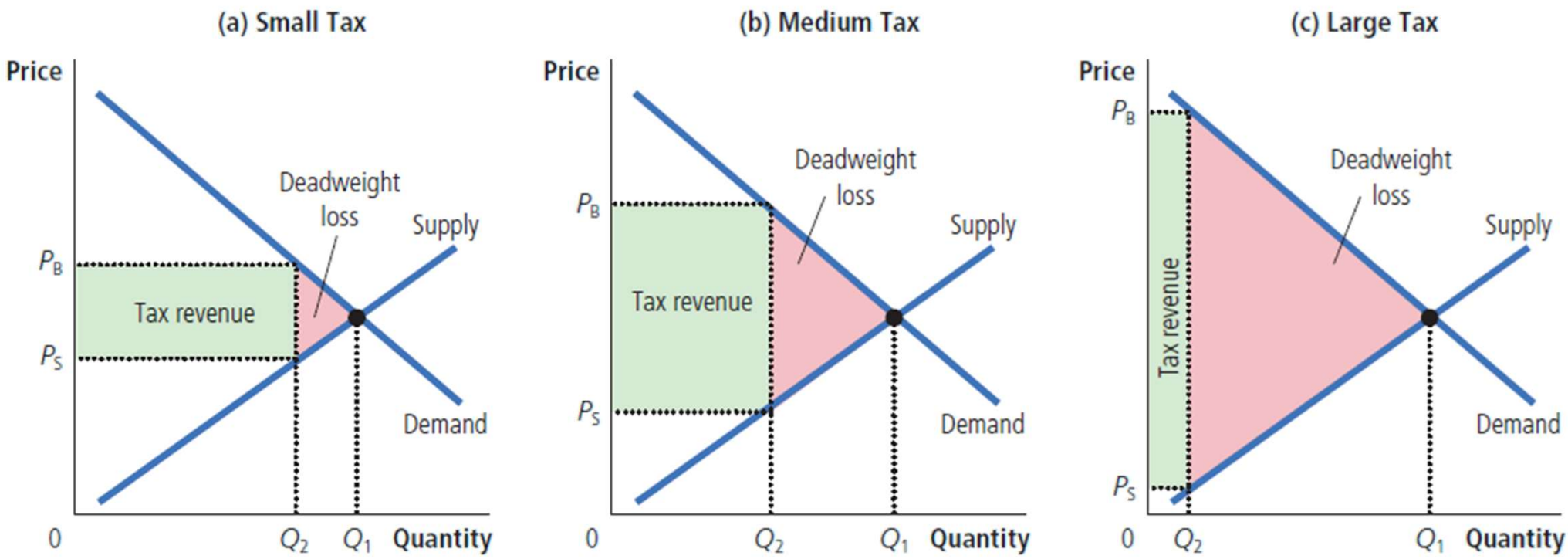
5.The obstacles to efficiency

- If supply is less elastic than demand, suppliers will bear a higher burden



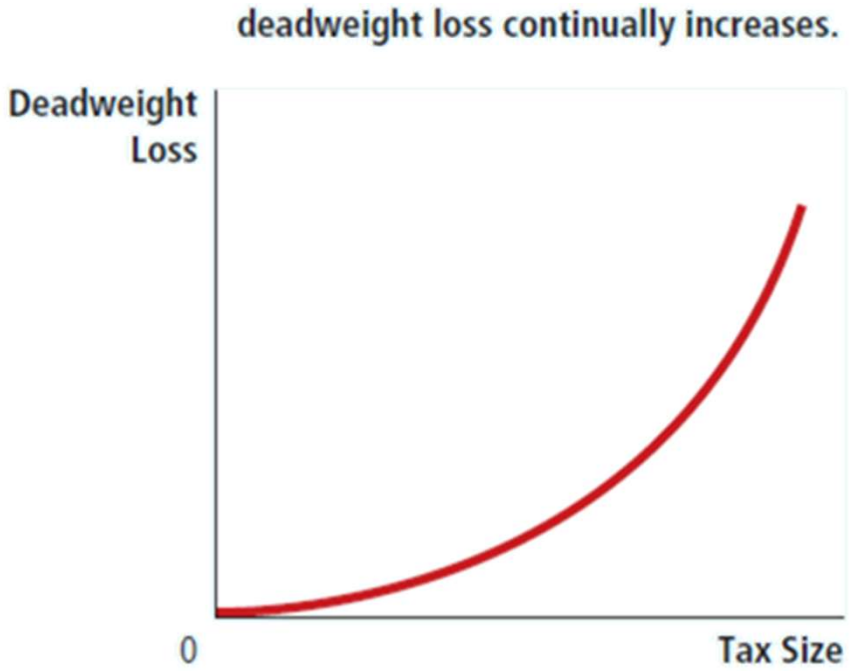
5.The obstacles to efficiency

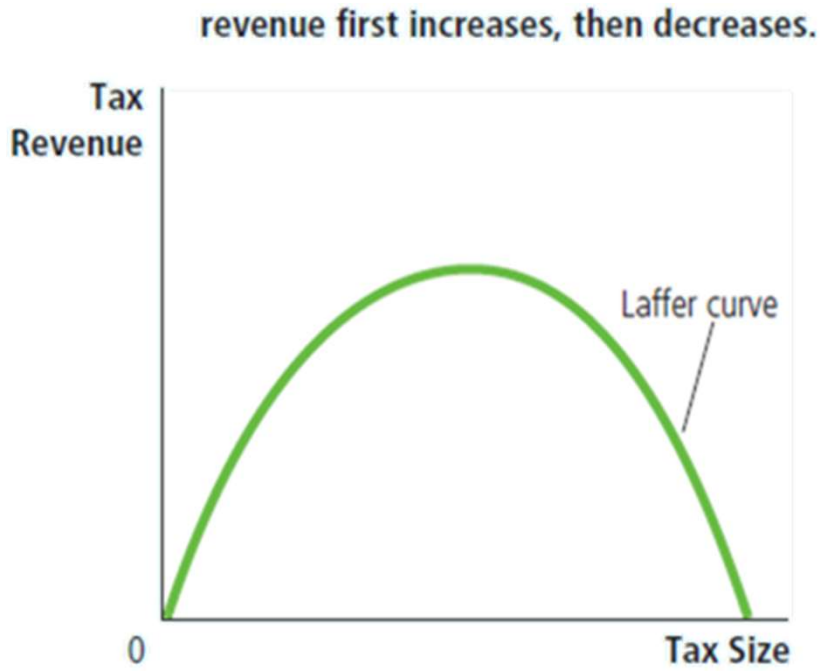
- Deadweight loss and tax revenue:



5.The obstacles to efficiency

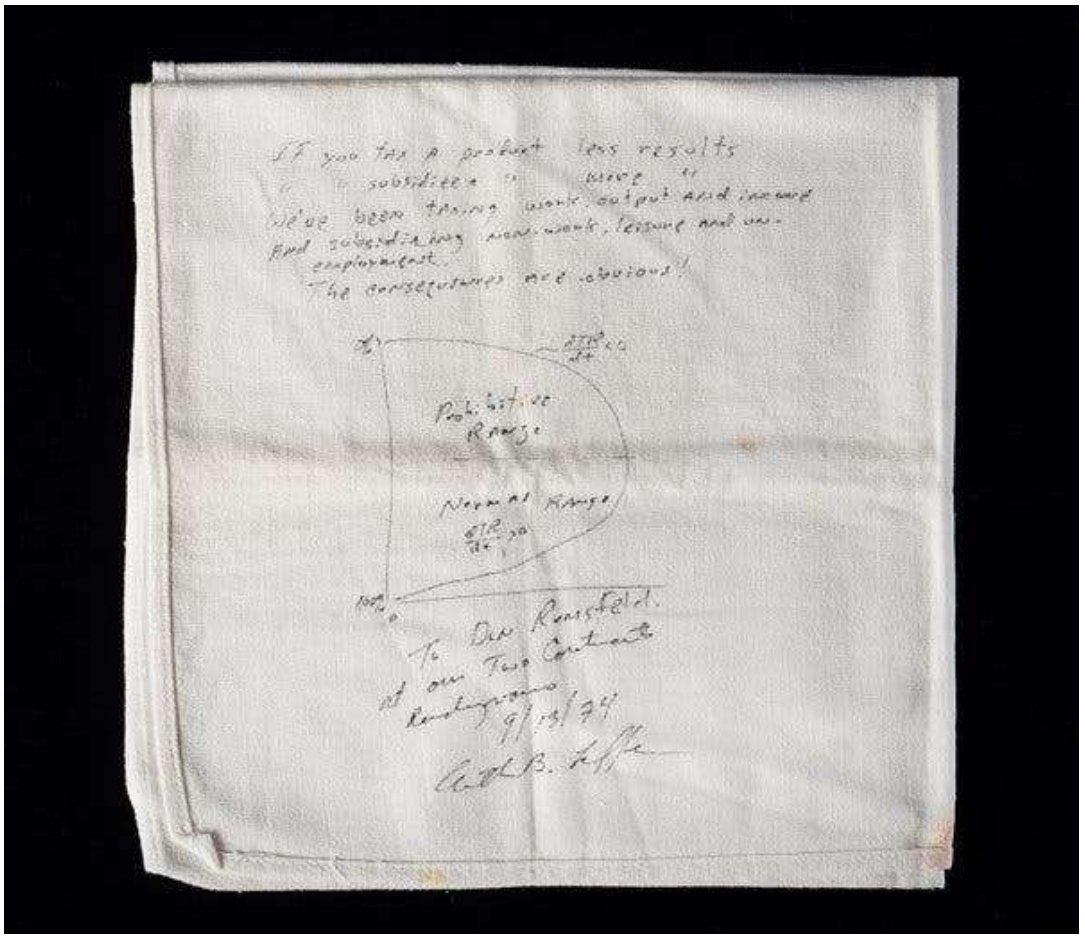
- Laffer Curve: Lower tax rates may actually lead to higher tax revenues (and vice versa)





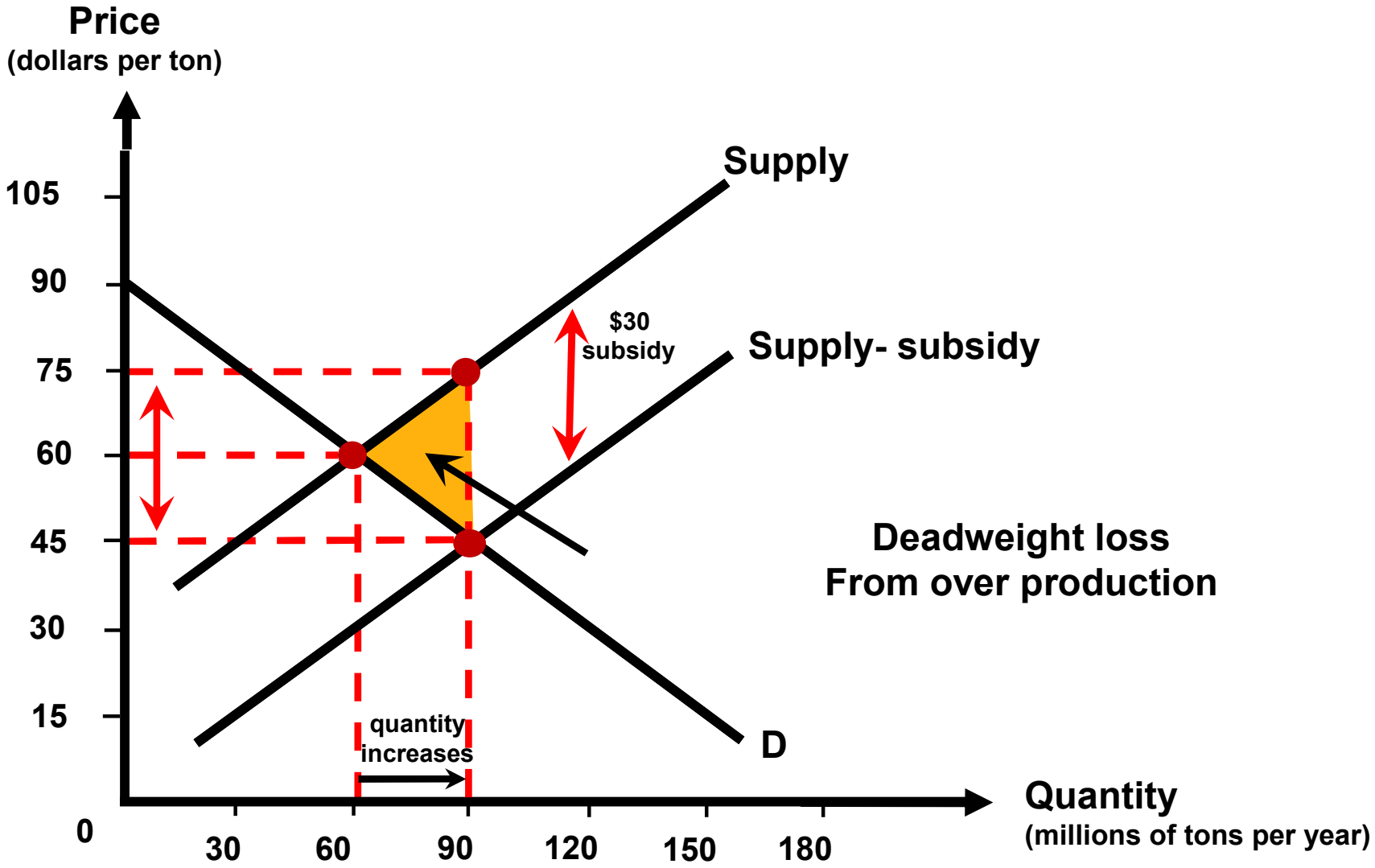
5.The obstacles to efficiency

- Laffer Curve was first introduced by Arthur Laffer in 1974:



5.The obstacles to efficiency

- Subsidies: payments made by the governments to producers



5.The obstacles to efficiency

- Production quota: imposing an upper limit on the quantity of a good that maybe produced

